

Université de Montréal

**Retrieval Augmented Generation in Financial QA: A
Systematic Study**

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Mémoire présenté en vue de l'obtention du grade de
Maître ès sciences (M.Sc.)
en Intelligence Artificielle

April 26, 2026

Université de Montréal

Faculté des arts et des sciences

Ce mémoire intitulé

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Résumé

Cette thèse étudie la recherche d'information dans les systèmes de génération augmentée par récupération appliqués aux questions financières. Dans ce domaine, les réponses doivent être fondées sur des documents primaires, souvent des rapports SEC longs, répétitifs et riches en tableaux. Le travail montre que la principale difficulté n'est pas seulement de retrouver le bon document, mais de trouver la bonne page à l'intérieur de ce document. Nous proposons d'abord un cadre d'évaluation avec conditions oracles qui sépare les erreurs au niveau du document, de la page et du passage. Appliqué à FinanceBench, ce cadre montre un écart important entre la reconnaissance du document et la reconnaissance de la page.

Pour réduire cet écart, la thèse introduit un pipeline hiérarchique qui classe d'abord les pages avec un bi-encodeur à paramètres partagés, puis effectue la recherche de passages seulement dans les pages les mieux classées. Le modèle est adapté au domaine financier et améliore le rappel des pages ainsi que la qualité des réponses numériques. La thèse présente aussi une extension de benchmark basée sur les questions LoFIN/HiREC provenant de FinQA et SEC-QA. Seule la partie FinQA a pu être alignée à des pages PDF, car les données SEC-QA disponibles ne contenaient pas de texte d'évidence suffisant pour faire la correspondance avec une page de référence. Enfin, nous analysons l'effet du découpage en segments et comparons l'extraction de texte avec une approche visuelle de type ColPali.

Mots-clés : génération augmentée par récupération, question-réponse financière, FinanceBench, FinQA, recherche hiérarchique, rapports SEC, PDF.

Abstract

This thesis studies retrieval-augmented generation for financial question answering over financial documents. In this setting, answers must be grounded in primary-source documents, and the main practical challenge is often retrieval. The system must find a specific page or table inside a long, repetitive filing before a language model can answer reliably. The thesis first introduces an oracle-based evaluation framework that decomposes retrieval failure into document-level, page-level, and chunk-level components. Applied to FinanceBench, the analysis shows that current systems can often identify the correct filing but still miss the evidence page, making within-document retrieval the main bottleneck.

To address this bottleneck, the thesis proposes a hierarchical page-then-chunk retrieval pipeline. A fine-tuned shared-parameter bi-encoder first scores pages for relevance, and standard dense chunk retrieval is then restricted to the highest-scoring pages. This design reduces noise from irrelevant sections and improves page recall and downstream numerical answer accuracy. The thesis also documents a benchmark-extension pipeline based on the LoFIN/HiREC benchmark, which is composed from FinQA and SEC-QA. The FinQA portion was aligned to PDF pages using fuzzy evidence matching, while the SEC-QA portion could not be added because no gold page text was available to map questions to gold PDF pages. In addition, the thesis evaluates chunking choices, compares PDF text extraction methods, and explores visual page retrieval with ColPali as an alternative to text-only retrieval.

Keywords: retrieval-augmented generation, financial question answering, FinanceBench, FinQA, hierarchical retrieval, SEC filings, PDF retrieval.

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List of acronyms and abbreviations

BCE	Binary Cross-Entropy loss
BM25	Best Match 25, a sparse term-based retrieval scoring function
DPR	Dense Passage Retrieval
EDGAR	Electronic Data Gathering, Analysis, and Retrieval system of the SEC
FT	Fine-Tuned
HyDE	Hypothetical Document Embeddings
LLM	Large Language Model
MNR	Multiple Negatives Ranking loss
NLP	Natural Language Processing
QA	Question Answering

RAG	Retrieval-Augmented Generation
RRF	Reciprocal Rank Fusion
SEC	U.S. Securities and Exchange Commission
VLM	Vision-Language Model

Acknowledgements

First and foremost, I would like to express my deepest gratitude to my supervisor, Philippe Langlais, for his support and guidance throughout this journey. His flexibility, patience, and insightful advice were key in bringing this work to life.

My time at the University of Montréal would not have been the same without the vibrant academic environment of the RALI lab. To all the lab members: thank you for the stimulating discussions, the shared insights, and the camaraderie that made the daily research process so much more rewarding.

Introduction

The financial industry runs on documents. Each quarter, thousands of publicly traded companies file detailed disclosures, accumulating a vast record of annual reports, quarterly filings, earnings transcripts, and special notices that constitutes the primary evidence base for analysts, investors, and compliance professionals. Answering precise questions about these filings (documents) is not only useful but essential to work in finance, such questions range from what a company's revenue was in a given fiscal year, to how management characterized a particular risk exposure, to what the terms of a recent acquisition were. Until recently, answering such questions required a human reader to navigate these documents manually, a process that is time-consuming, expensive, and prone to the errors that come with searching through hundreds of dense pages under pressure.

Large language models have recently introduced a new possibility, namely a system that reads a question in natural language and produces an answer drawn directly from the relevant context. The paradigm that makes this practical is retrieval-augmented generation (RAG), which connects the reasoning and fluency of a generative model with a retrieval component that identifies the most relevant passages from an external document corpus. The appeal for financial applications is considerable. Such a system can ground its answers in primary sources, support citation and auditability, and remain current as new filings are published without requiring the underlying language model to be retrained. Yet the gap between this description and reliable system in practice is substantial, and understanding it requires examining the retrieval component as well as the generation component.

The difficulty is not primarily one of language generation. Modern generative models are remarkably capable of synthesizing financial information when given the right context, and we believe that the primary challenge is retrieval. The system must first locate, within a corpus that may contain hundreds of lengthy filings, the specific page and passage containing the evidence needed to answer a given question. Financial filings are not only long (an annual report can exceed two hundred pages) but structurally repetitive. Revenue is discussed in the management commentary, in the income statement, in segment breakdowns, and in the revenue recognition footnote. The same section headers appear across companies and across

years. Dense tables present critical numerical facts in layouts that standard text extraction handles poorly. A retrieval system must navigate these challenges to surface not just a topically relevant passage but the specific context a particular question requires to generate an adequate answer.

This thesis takes the retrieval challenge as its central object of study. Its starting point is an observation that has received limited (systematic) attention in the literature. Namely, retrieval failure in long-document financial question answering is not a single phenomenon. A pipeline can fail because it does not retrieve the correct document, or because it retrieves the correct document but misses the relevant page within it, or because it locates the correct page but still fails to surface the most informative passage. Each failure mode has a different cause and different implications for how a system should be improved.

The first contribution of this thesis is an oracle-based evaluation framework that decomposes retrieval failure across multiple levels of granularity. By restricting the search space to the correct filing (oracle document) or to the correct pages within it (oracle page), the framework establishes empirical upper bounds at each level and measures the gap between current systems and those bounds. The framework further quantifies generative performance by measuring the maximum answer accuracy gain that better page retrieval alone can deliver, with the generator held fixed. Applied to FinanceBench [Islam et al., 2023], the primary benchmark for PDF-native financial question answering, this analysis reveals a consistent finding across retrieval methods spanning sparse, dense, hybrid, and hierarchical approaches. While current systems achieve reasonably high success at identifying the correct filing, their ability to find the relevant page within that filing falls substantially short of the theoretical ceiling. Thus the dominant bottleneck is within-document page retrieval, not document discovery.

The second contribution attempts to address this bottleneck directly. This thesis introduces a hierarchical page-then-chunk retrieval pipeline centred on a learned page scorer. In Stage 1, a bi-encoder fine-tuned via cross-validation on FinanceBench page-level relevance selects the top- P most relevant pages from the full corpus. In Stage 2, standard dense chunk retrieval is restricted to chunks from those pages, reducing noise from irrelevant filing sections while preserving fine-grained selection within the most promising regions. The thesis further introduces a fine-tuned reranker, initialized from a general-purpose reranking checkpoint and trained on Out-Of-Distribution (OOD) FinQA [Chen et al., 2021] financial page data, which is then applied zero-shot to the held-out FinanceBench benchmark. Together, these two components produce the strongest retrieval configuration evaluated in this thesis.

The third contribution is a systematic evaluation of retrieval and generation performance across multiple dimensions. On the retrieval side, all configurations are evaluated at multiple

levels of granularity, document recall, page recall, and context quality, and broken down by question type and filing type to surface where different methods succeed and fail. A qualitative failure analysis then examines the recurring reasons behind retrieval failures. On the generation side, the thesis evaluates answer quality across generator sizes and question types, and analyses the residual errors that persist even under perfect retrieval conditions.

The fourth contribution broadens the scope of evaluation through a benchmark extension pipeline. The thesis develops a procedure for aligning questions from the LoFIN/HiREC benchmark [Choe et al., 2025], which is composed of two existing financial question answering (QA) datasets, to gold PDF pages in SEC filings retrieved from EDGAR. The aligned result serves as a starting point toward a larger evaluation set that goes beyond FinanceBench’s 150 questions, and offers a preliminary test of whether retrieval performance holds on a separate question source. The scope of the current alignment is however limited by practical constraints discussed in the body of the thesis, and does not yet achieve the document and question type diversity that motivates the extension.

The fifth contribution explores visual document retrieval using ColPali [Faysse et al., 2024], which processes document pages as images rather than extracted text. This is evaluated as a potential path around the text extraction bottleneck that affects dense tables and complex layouts in financial filings. ColPali is evaluated both as a standalone retrieval system and as a reranking signal over dense text candidates, providing informative evidence about the conditions under which visual signals are advantageous for financial documents.

Chapter 1

Literature Review

1.1. Introduction

This chapter surveys the research landscape underpinning the contributions of this thesis. The narrative follows the logical progression of the problem itself, beginning with retrieval-augmented generation as a paradigm, tracing where it came from, how it has evolved, and where it currently stands. It then turns to the information retrieval methods that form the technical foundation of retrieval-augmented generation systems, examining their strengths and failure modes with the financial retrieval problem in mind. From there the focus narrows to financial question answering as an application domain, characterizing the properties of financial documents that make retrieval especially challenging and surveying the benchmarks that the field has developed to measure progress. The chapter then examines strategies for retrieval over long, structured documents, surveying hierarchical retrieval approaches from general-purpose multi-stage indexing systems to domain-specific financial pipelines, and treating chunking as a deliberate design variable rather than a fixed preprocessing choice. It closes with treatments of domain adaptation for financial retrieval and visual document retrieval as an emerging alternative paradigm.

1.2. Retrieval-Augmented Generation

1.2.1. From Parametric Knowledge to Grounded Generation

Large language models encode a remarkable breadth of world knowledge in their parameters, acquired through exposure to vast text corpora during pre-training [Brown et al., 2020]. Yet this parametric approach to knowledge carries inherent limitations that become acute in high-stakes application domains. A model’s knowledge is frozen at the time of its training and grows stale as the world changes. It cannot easily cite the specific sources that underlie

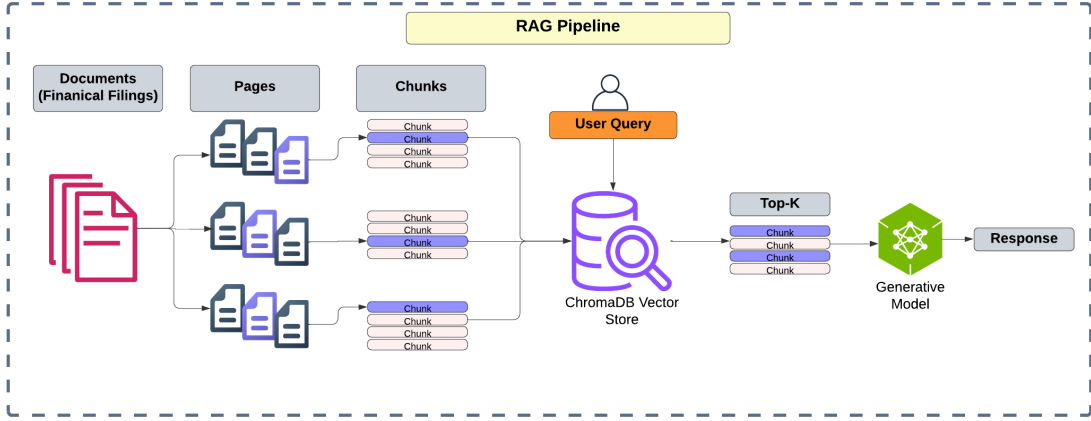


Fig. 1.1. Standard retrieval-augmented generation (RAG) pipeline. A query is used to retrieve relevant document fragments from an external corpus, which are then provided as context to a generative model to produce an answer.

any given claim, making it difficult to verify or audit, and because the model must compress everything it has learned into a fixed parameter count, it is prone to confidently generating plausible but factually incorrect statements, a failure mode commonly called hallucination, particularly when asked about facts that are rare, recent, or domain-specific [Brown et al., 2020, Ji et al., 2023]. In professional domains such as finance, where users need answers they can verify against primary sources and where a subtly incorrect number can have material consequences, these limitations are not only inconvenient but also unacceptable.

Retrieval-augmented generation (RAG) addresses these limitations by separating the language model’s role as a generator and reasoner from the task of knowledge storage and information retrieval. Figure 1.1 illustrates the standard RAG pipeline and highlights the central role of retrieval, whose effectiveness ultimately constrains the quality of generation. Rather than asking the model to recall facts from its parameters, a RAG system maintains an external corpus of documents that can be updated, versioned, and inspected independently. At inference time, a retrieval component identifies the most relevant passages from this corpus and supplies them to the language model as context, after which the model synthesizes those passages into a coherent answer. This architecture reduces hallucination by anchoring generation in retrieved evidence, enables provenance by allowing the retrieved passages to be surfaced alongside the answer, and supports knowledge currency by allowing the document corpus to be updated without retraining [Lewis et al., 2020]. The formal framework can be expressed concisely. Let $\mathcal{D} = \{d_1, \dots, d_{|\mathcal{D}|}\}$ be a corpus of documents, and let $\mathcal{C}$ denote the set of all textual chunks obtained by segmenting the documents in $\mathcal{D}$. Given a query q , a

retrieval function R_k selects the k most relevant chunks according to a scoring function s ,

$$R_k(q) = \text{Top-}k_{c \in \mathcal{C}} s(q, c), \quad (1.2.1)$$

and a generator G_θ produces an answer conditioned on the query and the retrieved context,

$$\hat{y} = G_\theta(q, R_k(q)). \quad (1.2.2)$$

In this formulation, the quality of $\hat{y}$ is bounded by the quality of $R_k(q)$. If the relevant evidence is not retrieved, even the most capable generator cannot produce a correct, grounded answer. This tight coupling between retrieval quality and generation quality is the central observation motivating the present thesis.

1.2.2. The Origins of RAG and Open-Domain QA

The formal introduction of the RAG framework is typically attributed to Lewis et al. [2020], who combined a pre-trained neural retriever with a sequence-to-sequence generator and demonstrated substantial improvements over closed-book baselines on knowledge-intensive natural language processing tasks. Their formulation treats retrieved passages as a latent variable that the generator marginalizes over, establishing a probabilistic grounding for the retrieve-then-generate pipeline that subsequent work has largely preserved in spirit even while modifying its implementation substantially.

This architecture built directly on the work of open-domain question answering research that had already demonstrated the value of retrieval-based approaches [Chen et al., 2017, Lee et al., 2019]. What the integration with large pre-trained language models added was a qualitative shift in generation quality. Rather than extracting a span from a single retrieved passage, the model could synthesize information from multiple passages, paraphrase, aggregate, and reason in ways that earlier extractive systems could not. Fusion-in-Decoder [Izacard and Grave, 2021] extended this insight by encoding each passage separately and fusing them in the decoder’s attention, enabling more robust multi-passage reasoning and providing a strong generation backbone that many subsequent RAG systems have built upon.

1.2.3. The Evolution of RAG Paradigms

The simplest implementations of retrieval-augmented generation, which the survey paper from Gao et al. [2023b] characterise as Naive RAG, follow a three-stage pipeline. Documents are indexed offline as dense vector representations, a query is encoded at inference time and matched to the nearest indexed vectors, and the top- k retrieved passages are concatenated with the query and passed to the language model for generation. This design is appealing for its simplicity and modularity, and it achieves strong results on general-domain benchmarks

where queries are short, documents are compact, and relevant passages are relatively easy to distinguish from irrelevant ones.

In more demanding settings, the limitations of this approach become apparent. Chunking documents into fixed-size windows can fragment evidence that spans multiple passages, leaving no single chunk containing all the information the generator needs. Retrieving by vector similarity favours passages that are topically related to the query but may not contain the specific evidence it requires, a problem that grows more severe as documents become longer and more internally repetitive. And because the retriever and generator are trained independently, the retriever has no direct signal about which passages are actually useful for downstream generation. These limitations set the research grounds for what Gao et al. [2023b] call Advanced RAG.

Advanced RAG systems introduced targeted interventions at both the beginning and the end of the retrieval process. Pre-retrieval optimization focuses on improving the match between the query and the indexed passages before any retrieval takes place. Query reformulation methods expand or decompose the original question, they can generate a hypothetical answer document whose embedding may align better with relevant passages, or break a complex question into simpler sub-questions that can each be matched to a specific passage [Gao et al., 2023a, Ma et al., 2023, Karpukhin et al., 2020]. Index optimization techniques restructure the document representation to make retrieval more reliable, for instance by preserving table structure, encoding section metadata, or building hierarchical indices that capture document organization [Gao et al., 2023b].

Post-retrieval optimization intervenes after the initial candidate set has been assembled. Reranking passes retrieved candidates through a more powerful but computationally expensive scoring model that can evaluate query-passage relevance more accurately than the first-stage retriever [Nogueira et al., 2019]. Context compression removes passages judged redundant or irrelevant, allowing the generator to focus on the most pertinent evidence [Xu et al., 2023].

More recent developments have moved toward what Gao et al. [2023b] describe as Modular RAG, in which retrieval and generation components are decomposed into independently configurable modules that can be orchestrated in diverse patterns depending on the task. Iterative retrieval allows the system to perform multiple rounds of retrieval interleaved with partial generation, using the model’s intermediate outputs to guide subsequent queries [Shao et al., 2023]. Adaptive retrieval enables the system to decide based on the model’s assessed uncertainty whether a given question requires retrieval at all or can be answered from parametric knowledge [Jiang et al., 2023]. The current state-of-the-art (SOTA) is one in which the core retrieve-then-generate pipeline is well understood and research attention has shifted

toward specialised domains, improved diagnostic evaluation, and the identification of the specific failure modes that aggregate metrics conceal.

1.3. Information Retrieval Methods for RAG

The practical performance of a RAG system is determined in large part by the quality of its retrieval component. This section surveys the main classes of retrieval method that appear in the RAG literature and in the experimental evaluation of this thesis, examining their respective strengths and failure modes with the financial retrieval problem specifically in view.

1.3.1. Sparse Retrieval

Sparse retrieval methods represent documents and queries as high-dimensional sparse vectors in vocabulary space, where each dimension corresponds to a term in the vocabulary and the vast majority of values are zero. The most widely deployed sparse retrieval function is BM25 [Robertson and Zaragoza, 2009b], which scores a document D against a query Q as

$$\text{BM25}(D, Q) = \sum_{t \in Q} \text{IDF}(t) \cdot \frac{f(t, D) (k_1 + 1)}{f(t, D) + k_1 \left(1 - b + b \frac{|D|}{\text{avgdl}} \right)}, \quad (1.3.1)$$

where $f(t, D)$ is the frequency of term t in document D , $|D|$ is the document length, avgdl is the average document length in the collection, and k_1 and b are tuning parameters controlling term frequency saturation and length normalization respectively. The inverse document frequency term $\text{IDF}(t)$ rewards terms that are distinctive across the corpus, ensuring that rare terms carry more weight than common ones.

BM25 remains a competitive baseline in many specialised retrieval settings because its reliance on exact term matching is precisely what makes it effective when the query and the relevant document share distinctive surface forms. Named entities, technical identifiers, ticker symbols, and fiscal year markers are all retrieved reliably if the query contains them verbatim. BM25 is also transparent (its scores decompose directly to show which query terms contributed most to a document’s ranking) and efficient, relying on inverted indices that support fast retrieval over large corpora [Lin et al., 2021]. Its limitation is the complementary side of its strength. The method cannot account for semantic equivalence, so a query about “automobile manufacturing revenue” will not reliably retrieve a passage written in terms of “vehicle production sales” unless there is meaningful lexical overlap.

Learned sparse retrieval methods such as SPLADE [Formal et al., 2021] attempt to address vocabulary mismatch while preserving the practical advantages of sparse indexing. Using a transformer-based encoder, SPLADE assigns vocabulary-space weights to queries

and documents based on contextualized representations rather than raw term frequencies, effectively performing implicit query expansion. The result is a sparse representation that retains compatibility with inverted index retrieval while capturing some semantic generalization that pure lexical methods lack.

1.3.2. Dense Retrieval and the Bi-Encoder Architecture

Dense retrieval methods represent queries and documents as compact, continuous vectors in a shared embedding space, enabling retrieval based on semantic similarity rather than surface-level term overlap. The dominant architecture is the bi-encoder, which encodes queries and documents independently into fixed-dimensional dense representations and scores relevance as their cosine similarity. In many modern retrieval models, including the setup used in this thesis, the same parameterized encoder is applied to both the query and chunk inputs,

$$s(q, c) = \frac{\langle \mathbf{h}_q, \mathbf{h}_c \rangle}{\|\mathbf{h}_q\| \|\mathbf{h}_c\|}, \quad \mathbf{h}_q = f_\theta(q), \quad \mathbf{h}_c = f_\theta(c). \quad (1.3.2)$$

Because document embeddings can be precomputed and indexed offline, retrieval reduces to a nearest-neighbour search in embedding space, which is fast and scalable even over large corpora. Models such as DPR [Karpukhin et al., 2020] and BGE-M3 [Chen et al., 2025] demonstrated that pre-training on large-scale data followed by contrastive fine-tuning on question-passage pairs produces embeddings substantially more effective for question answering retrieval than sparse baselines on general-domain benchmarks.

The key advantage of the formulation in Equation (1.3.2) is its ability to capture semantic similarity without requiring shared vocabulary. A query about “profitability” can match a passage discussing “net income margins” even if neither term appears in the other. This semantic generalization makes dense retrievers particularly effective for queries phrased in natural conversational language.

The contrastive training objective that underlies most dense retrieval models teaches the encoder to place queries close to their relevant passages and far from irrelevant ones in embedding space. Negative examples play a critical role in shaping what the model learns. Easy negatives drawn at random provide weak signal, while hard negatives drawn from topically similar but answer-irrelevant passages teach the model finer-grained distinctions. This distinction between easy and hard negatives becomes especially important for domain adaptation, as discussed in Section 1.7.

1.3.3. Hybrid Retrieval and Reciprocal Rank Fusion

Sparse and dense methods exhibit complementary failure modes. Sparse methods fail when query and document vocabulary diverge, but succeed when queries contain rare or

distinctive terms that must appear verbatim. Dense methods succeed at semantic matching but can be distracted by topically related passages that do not contain the specific evidence required. Hybrid retrieval combines both to exploit their complementarity, typically by running sparse and dense retrievers independently and fusing their ranked lists [Luan et al., 2021].

Reciprocal Rank Fusion [Cormack et al., 2009] provides a simple and robust fusion strategy that focuses on rank positions rather than absolute scores, which are on incomparable scales across retrievers,

$$\text{RRF}(d) = \sum_{r \in \text{rankers}} \frac{1}{k_{\text{rrf}} + \text{rank}_r(d)}, \quad (1.3.3)$$

where $\text{rank}_r(d)$ is the position of document d in the ranking from retriever r , and k_{rrf} (typically set to 60) prevents small rank differences from dominating the score. In financial document retrieval specifically, hybrid approaches are particularly valuable because sparse retrieval reliably captures company names, ticker symbols, and fiscal year markers, while dense retrieval handles semantic variations in question phrasing.

1.3.4. Query Reformulation and Hypothetical Document Embeddings

A structural challenge in retrieval for question answering is the asymmetry between the short, high-level phrasing of a natural language question and the long, detailed phrasing of a relevant passage. A query of the form “What was the company’s FY2022 net revenue?” may share few tokens with the income statement row it is trying to locate. Query reformulation methods address this asymmetry by transforming the query before retrieval to better match the style and vocabulary of relevant passages.

Hypothetical Document Embeddings (HyDE) [Gao et al., 2023a] address this problem directly. Rather than embedding the question itself, HyDE prompts a language model to generate a hypothetical passage that would answer the question, then uses that hypothetical passage’s embedding for retrieval. The generated passage does not need to contain accurate facts, it simply needs to be phrased in a style and vocabulary that aligns with the documents in the corpus. For financial queries, this means the hypothetical passage is written in the tone of a financial report, with appropriate terminology and sentence structures, which can substantially improve the match between the query embedding and the embeddings of genuine evidence passages. Multi-HyDE, a variant evaluated in this thesis, averages the embeddings of multiple independently sampled hypothetical passages to reduce variance and

improve retrieval stability, as expressed by

$$\mathbf{h}_{\text{Multi-HyDE}}(q) = \frac{1}{N} \sum_{n=1}^N g_{\theta}(\hat{d}_n), \quad (1.3.4)$$

where $\hat{d}_1, \dots, \hat{d}_N$ are independently sampled hypothetical passages generated by the language model given query q , and f_{θ} is the shared encoder used by the bi-encoder.

1.3.5. Reranking with Cross-Encoders

First-stage retrieval with bi-encoders is fast and scalable but limited in its ability to model fine-grained query-document interactions, because each input is encoded independently. Cross-encoder rerankers address this by jointly encoding the query and candidate passage through a transformer that attends to both simultaneously. For a cross-encoder with parameters ϕ , the relevance score for a query q and candidate passage c is

$$r_{\phi}(q, c) = f_{\phi}([\text{CLS}] \ q \ [\text{SEP}] \ c \ [\text{SEP}]), \quad (1.3.5)$$

where f_{ϕ} is a transformer that operates over the full concatenated sequence and produces a scalar relevance estimate. This enables the model to capture nuanced relevance signals, such as matching a specific fiscal year in the query to a year in the passage header, or recognizing that a passage discusses a related metric rather than the specific one asked about, in ways that a bi-encoder cannot easily learn. Because cross-encoders require a full forward pass for each query-document pair, they are too expensive for first-stage retrieval over large corpora but well-suited as a second stage that reranks a small set of candidates produced by a fast first-stage retriever [Nogueira et al., 2019].

In financial documents, rerankers are particularly valuable for resolving near-duplicate sections. Different fiscal years’ financial statements follow identical formats with nearly identical language, and cross-encoder rerankers can exploit subtle contextual cues to make the fine-grained distinctions that bi-encoders miss.

1.3.6. Late Interaction Models

A middle path between the efficiency of bi-encoders and the expressiveness of jointly-encoding cross-encoders is represented by late interaction architectures, of which ColBERT [Khattab and Zaharia, 2020] is the most prominent. Rather than condensing the entire document into a single vector, ColBERT produces a sequence of contextualized token embeddings for each document. Relevance is then computed via the MaxSim operator, where for each query token embedding $\mathbf{e}_i^q$ the maximum cosine similarity to any document token embedding

$\mathbf{e}_j^d$ is taken, and these per-token maxima are summed over all query tokens,

$$S_{\text{ColBERT}}(q, d) = \sum_{i=1}^{|q|} \max_{j=1}^{|d|} \cos(\mathbf{e}_i^q, \mathbf{e}_j^d). \quad (1.3.6)$$

Document token embeddings can be precomputed offline, so retrieval remains efficient despite the richer interaction model. Equation (1.3.6) is the conceptual foundation for ColPali, the visual retrieval model discussed in Section 1.6.

1.4. Financial Question Answering

1.4.1. The Distinctive Challenges of Financial Documents

Financial question answering is a demanding application domain for retrieval systems because it requires searching over documents that exhibit multiple challenging properties, both individually and in combination. A typical 10-K annual report spans between one hundred and three hundred pages and follows a standardized structure mandated by regulation, covering the business description, risk factors, financial statements, management’s discussion and analysis, and a collection of exhibits [Islam et al., 2023]. This standardization means that similar section headers appear across every company and every fiscal year, creating a retrieval challenge distinct from that of a heterogeneous document corpus. A query about operating expenses is relevant to the income statement, but the income statement in one year’s 10-K is structurally nearly identical to the income statement in every other year’s 10-K from every other company. Retrieval systems relying on topical similarity must therefore make fine-grained distinctions that first-stage bi-encoders frequently fail to make reliably.

Financial documents are also numerically dense in a way that matters for question answering. Many of the most important questions that analysts care about, including revenue figures, earnings per share, capital expenditures, and debt ratios, require extracting or computing specific numerical values from dense tabular contexts [Chen et al., 2021]. These values appear as entries in tables whose structure encodes essential context. The column header identifies what is being measured, the row label identifies the entity or period, and the cell value is the answer.

Finally, financial questions require precision in a sense that general-domain questions do not. A question asking for FY2022 revenue must be answered with the value from the 2022 filing, not the 2021 or 2023 filing, even though all three filings are from the same company and discuss the same concept in nearly identical language. This temporal specificity imposes a retrieval requirement that goes beyond topical relevance and extends to exact contextual matching of year references, quarter identifiers, and other temporal markers. Combined with the structural repetitiveness described above, this creates a retrieval setting in which passages

from different reporting periods are often semantically and structurally near-identical. As a result, retrieval is not simply a matter of identifying the correct topic or year, but of resolving fine-grained distinctions within highly similar content, making accurate ranking non-trivial.

1.4.2. The Evolution of Financial QA Benchmarks

The benchmark landscape for financial question answering reflects a progression from closed-book settings, where the relevant document or passage is provided to the model, toward open-book settings where the model must retrieve its own evidence from a realistic document corpus. This progression matters for interpreting what different benchmarks measure and what limitations they impose.

The earliest influential benchmarks in the space, including FinQA [Chen et al., 2021] and TAT-QA [Zhu et al., 2021], focused primarily on numerical reasoning capability. Both datasets provide question-answer pairs grounded in short document contexts that contain the necessary evidence, essentially supplying the retrieval result to the model and asking only whether it can reason over what has been handed to it. FinQA in particular annotates reasoning programs that trace the multi-step calculations required to derive each answer, making it a valuable resource for studying arithmetic reasoning. ConvFinQA [Chen et al., 2022] extended this framework to conversational settings, introducing multi-turn dependencies where each question builds on the previous one. These benchmarks established that language models can perform sophisticated numerical reasoning over financial tables and narrative text, but because they supply the evidence rather than requiring its retrieval, they do not measure the retrieval component of financial QA.

More recent benchmarks have introduced retrieval requirements in varying forms. SEC-QA [Lai et al., 2024] provides a systematic evaluation corpus for financial QA over SEC filings with an emphasis on cross-company comparisons, though its use of template-based question generation produces questions that may not capture the heterogeneous, narrative-driven nature of genuine analyst queries. LOFin [Choe et al., 2025] offers a large-scale open-domain benchmark with QA pairs spanning a very large number of SEC documents, enabling evaluation under conditions where the system must identify the relevant company and filing period before locating specific evidence.

1.4.3. FinanceBench as the Primary Evaluation Standard

Among existing benchmarks, FinanceBench [Islam et al., 2023] stands out as the most directly relevant to the work of this thesis. As far as we know, it is the only open-source available benchmark that requires end-to-end retrieval over real PDF filings with explicit

page-level evidence annotations, making it possible to evaluate retrieval quality at the granularity this thesis is most concerned with.

FinanceBench comprises question-answer-evidence triplets based on real SEC filings from forty U.S. publicly traded companies spanning nine GICS sectors. Each sample includes a natural language question, a reference answer, a verbatim evidence string extracted from the source document, and metadata identifying the source filing and the page number on which the evidence appears. This page-level annotation is what makes FinanceBench uniquely valuable, it provides ground-truth labels for page-level retrieval evaluation, enabling the oracle analysis framework described in Chapter 3 to operate at the granularity where the critical failure mode lies.

The questions in FinanceBench fall into three types, with fifty questions of each kind in the open-source subset. Domain-relevant questions ask about facts broadly applicable across companies, such as dividend policies or executive compensation structures, and often require locating narrative statements that may appear in varying sections of different filings. Metrics-generated questions require extracting or computing specific financial indicators from dense tabular contexts, these are especially demanding for retrieval because the answer lies in a specific table cell, and topically similar passages from adjacent pages or adjacent fiscal years serve as near-perfect retrieval decoys. Novel-generated questions are company-specific and require reasoning beyond simple extraction. The document types in the benchmark are dominated by 10-K annual reports at approximately 75%, with the remainder drawn from 10-Q quarterly reports, 8-K current reports, and earnings call transcripts.

1.4.4. Limitations of Existing Benchmarks and the Motivation for Extension

While FinanceBench provides an invaluable structured testbed, its 150-question open-source subset imposes real constraints on what can be concluded from experiments conducted on it alone. The strong skew toward 10-K filings means that findings about retrieval difficulty, chunking effects, or page scorer performance on annual reports may not generalize to quarterly reports, earnings transcripts, or more specialized filing types. The limited number of open-source questions means that stratified analysis by question type or filing type is based on small subgroups where individual question characteristics can have an outsized effect on aggregate metrics. Finally, the lack of separate train, dev, and test sets limits the generalization claims of domain fine-tuning for PDF-native benchmarks.

These limitations directly motivate the benchmark extension described in Section 3.6. At a broader level, they reflect a genuine gap in the financial QA literature. Most benchmarks that require retrieval either operate in a closed-book setting, assume that text has already

been extracted from the original document, have limited or no publically available data, or restrict evaluation to a single narrow filing type. A benchmark that stresses end-to-end PDF retrieval at scale across diverse filing types and question styles remains an open contribution, and the extension work of this thesis is a step toward filling it.

1.4.5. Recent RAG Systems for Financial QA

Several recent systems have pushed state-of-the-art performance on FinanceBench and related benchmarks, and understanding their contributions is important for contextualizing the contributions of this thesis. FinSage [Wang et al., 2025] introduces a multi-aspect RAG framework that combines domain-specific preprocessing, multi-path retrieval over different document representations, and a reranker fine-tuned using direct preference optimization on financial query-document pairs. The preprocessing pipeline is particularly thoughtful, preserving table structure, encoding section metadata, and constructing semantically coherent chunks that respect the document’s visual layout, an approach that the authors show improves retrieval precision substantially over naive fixed-size chunking.

Kim et al. [2025] propose a three-phase pipeline integrating query expansion, hybrid retrieval with fine-tuned embedders, and a DPO-trained reranker with document selection, achieving strong improvements in retrieval precision and downstream answer accuracy across several financial QA datasets. These systems demonstrate that substantial performance gains are achievable through careful engineering of the full RAG pipeline.

What neither system provides, however, is a systematic decomposition of where their improvements come from. Neither study separates document-level retrieval performance from page-level retrieval performance, nor establishes oracle bounds that quantify the headroom remaining at each level of the retrieval hierarchy. It is therefore difficult to know from their reported results whether further investment in retrieval quality would yield significant returns, or whether the remaining performance gap is attributable to generation limitations that better retrieval cannot address. The oracle-based analysis framework introduced in this thesis is designed to fill precisely this diagnostic gap.

1.5. Hierarchical Retrieval Strategies for Long Documents

A fundamental challenge in building retrieval systems for long documents is selecting the appropriate granularity for the units that will be indexed and retrieved. Retrieving at the level of full documents preserves all context but makes it impractical for the retrieval system to identify which part of a long filing contains the relevant evidence, forcing the

generator to process an enormous amount of irrelevant text. Retrieving small chunks enables precise localization but risks fragmenting evidence that spans multiple passages and loses the surrounding context that helps the generator interpret what it has found. Hierarchical retrieval approaches address this tension by organizing documents into multi-level structures, using coarser levels to quickly identify the most promising regions of the document space and finer levels to precisely localize answer-bearing evidence within those regions. This section surveys the main families of hierarchical retrieval method, from general-purpose recursive indexing schemes to specialized pipelines designed for long financial documents.

1.5.1. Multi-Stage Retrieval and Hierarchical Indexing

A prominent family of approaches addresses the granularity tension by constructing hierarchical document representations offline, which are then searched in a coarse-to-fine manner at retrieval time.

RAPTOR [Sarathi et al., 2024] constructs a tree of abstractive summaries by recursively clustering document chunks and generating summary nodes at increasing levels of abstraction. A leaf node represents a short passage, its parent summarizes a cluster of related passages, and the root node represents a high-level synopsis of the full document. At retrieval time, the system can be queried at any level of the tree or across multiple levels simultaneously, allowing high-level questions to be matched to high-level summaries and detailed questions to be matched to fine-grained passages. On long-document benchmarks, this design substantially improves recall on questions requiring the synthesis of information from multiple disparate sections, because the summarization step aggregates evidence that no individual passage contains. The principal limitation is that recursive summarization can introduce information loss and hallucination at higher levels of abstraction, making the quality of the intermediate summaries a critical determinant of overall retrieval performance.

More generally, two-stage coarse-to-fine pipelines interleave a cheap first-stage retriever over coarse units (documents or sections) with a more expensive second-stage ranker over fine-grained units (passages or chunks) within the top- M coarse candidates. This design concentrates the second stage on a small, pre-filtered candidate pool, allowing a computationally expensive ranker to operate efficiently while covering the full document corpus at the first stage. The architecture introduces a new failure mode, where any relevant passage whose parent document or section is not retrieved at the first stage cannot be recovered, making the coarse retrieval component a hard ceiling on overall recall.

1.5.2. Parent-Child Structures and Decoupled Retrieval Granularity

A closely related pattern is the parent-child document structure, in which fine-grained child chunks are indexed for retrieval precision while coarser parent chunks are returned to the generator to provide broader context [Gao et al., 2023b]. After retrieving the top- k child chunks by similarity to the query, the system maps each child to its parent, deduplicates, and returns the unique parent chunks as the context for generation. This decoupling of retrieval granularity from generation granularity addresses a challenge, the optimal unit size for retrieval is often not the optimal unit size for generation, because retrieval benefits from short, discriminative passages while generation benefits from longer passages that supply complete context. In practice, a common instantiation uses sentences or short fixed-size spans as children and paragraphs or pages as parents, though the appropriate granularity depends heavily on the document type and the nature of the questions being asked.

The parent-child pattern is relevant to long financial documents because the evidence for a single question often occupies a coherent physical unit such as a financial statement page or a management discussion subsection, while precise retrieval requires matching the query against specific sentences or table rows within that unit. Indexing at both granularities, without committing to either, is a design principle that recurs across several recent financial retrieval systems.

1.5.3. Chunking as a Granularity Decision

Chunking sits at the boundary between document processing and retrieval design, and is most naturally understood as the decision that defines the leaf nodes of the retrieval hierarchy. Its importance is widely acknowledged, but it is rarely treated as a primary variable of study. Most financial RAG papers fix a chunking configuration early in system design and optimize other components without varying it, so the literature offers many data points about what retrieval performance is achievable with a given strategy but little evidence about the interactions between chunking choices and retrieval architecture.

The most common approach, fixed-size chunking, divides document text into overlapping windows of a fixed token count. Given a document of length n tokens, a sliding window of size w with overlap o (and stride $s = w - o$) produces $\lceil \frac{n-o}{w-o} \rceil$ chunks, where each chunk begins s tokens after the previous one. This approach requires no document understanding and is simple to scale, but it treats documents as undifferentiated character streams. In financial filings, where a table broken at a row boundary loses the correspondence between values and their column headers, these distortions can directly degrade retrieval quality.

Structure-aware chunking strategies attempt to respect the document’s natural organization. Section-boundary preservation prevents chunks from crossing the mandated section divisions of SEC filings. Table-aware splitting treats tabular regions as atomic units, keeping column headers co-located with the data rows they describe. Boilerplate detection identifies legal disclaimers and standardized caveats that can dilute retrieval quality if indexed without downweighting. Wang et al. [2025] demonstrate that structure-aware preprocessing substantially improves retrieval precision on FinanceBench relative to naive fixed-size chunking, with the largest gains on metrics questions where precise table retrieval is decisive. Yepes et al. [2024] similarly find that preserving table structure within chunk text improves downstream numeric answer accuracy.

The interaction between chunk size and retrieval architecture is the aspect of chunking that has received least systematic attention. Larger chunks preserve more co-located context, smaller chunks produce more discriminative embeddings that better distinguish between adjacent passages. This trade-off also depends on whether a hierarchical stage has already narrowed the candidate pool. When retrieval is restricted to a small set of pre-selected pages, the optimal within-page chunk size may differ from what is optimal for global search over an entire corpus. This dependence between chunking configuration and retrieval architecture motivates treating chunk size as a controlled experimental variable in this thesis, as described in Section ??.

1.5.4. Hierarchical Retrieval for Long Financial Documents

Financial QA introduces additional structure that specialized hierarchical systems can exploit. A typical end-to-end pipeline must navigate from a large corpus of filings (thousands of documents) to a single filing, then from that filing’s many pages to the specific page containing the answer, and finally to the precise passage or table row within that page. Treating these as separate retrieval problems rather than one allows each stage to be optimized for its specific discriminative challenge.

The LOFin benchmark and HiREC system provide a direct illustration of this principle for open-domain financial QA. LOFin is a large-scale evaluation corpus that requires a system to identify the relevant company and filing period from a collection of over 145,000 SEC documents before locating specific evidence. HiREC addresses this challenge with a hierarchical retrieval pipeline that first identifies related documents to resolve company and period ambiguity, and then selects the most relevant passages within those specific documents. This is followed by an evidence curation step that filters out irrelevant data and generates complementary queries if the initial retrieval is insufficient. The system’s explicit separation of the document retrieval process from the within-document passage selection

provides a principled way to reduce confusion caused by the repetitive boilerplate content and similar table structures inherent in standardized financial reports.

Within financial documents, the page provides a practical intermediate unit between the full filing and individual text chunks. Pages often correspond to coherent units of presentation, such as complete financial statements or self-contained subsections, meaning that their boundaries tend to align with meaningful semantic structure. In contrast, section boundaries require additional parsing and are not consistently defined across filings, while fixed-size chunking can arbitrarily fragment structured content. Because page boundaries are explicitly defined by the document format and consistently available across filings, they offer a stable and interpretable granularity for retrieval. More broadly, this reflects a general observation in retrieval systems that the choice of representation granularity can significantly impact retrieval effectiveness [Izacard and Grave, 2021, Liu et al., 2024].

1.6. Visual Document Retrieval

1.6.1. The Text Extraction Bottleneck

All the retrieval methods described in the preceding sections share a fundamental assumption, namely that documents can be adequately represented as sequences of text tokens. For financial filings, this introduces a systematic source of information loss. In practice, important aspects of these documents are encoded through visual structure rather than purely textual content. Layout plays a functional role, column alignment defines relationships between entities and values, and formatting cues such as headings and spacing convey hierarchical organization. Standard text extraction and parsing pipelines recover most narrative text reliably, but they can fail on complex multi-column tables, sometimes producing linearized representations in which values are separated from the headers that give them meaning [Xu et al., 2020]. For metrics-generated questions, whose answers often lie in these tabular structures, such extraction errors can propagate directly to retrieval, limiting the effectiveness of downstream models.

1.6.2. Vision-Language Models and the ColPali Architecture

Vision-language models offer an alternative approach that bypasses text extraction entirely by processing document pages as images. A vision encoder, typically a Vision Transformer that divides each page image into a grid of patches, represents the visual content of the page as a sequence of patch embeddings, preserving spatial structure rather than linearizing it [Dosovitskiy et al., 2020]. These patch embeddings can then be processed by

a language model backbone to produce contextualized representations that integrate visual layout with semantic content.

ColPali [Faysse et al., 2024] is the most directly relevant vision-language model for document retrieval. It combines the PaLIGemma architecture [Beyer et al., 2024] with the ColBERT late-interaction retrieval formulation, producing page-level retrieval by scoring relevance as

$$S_{\text{ColPali}}(q, p) = \sum_i \max_j \cos(\mathbf{e}_i^q, \mathbf{e}_j^p), \quad (1.6.1)$$

where $\mathbf{e}_j^p$ are the patch embeddings of page p produced by the vision encoder, and $\mathbf{e}_i^q$ are the query token embeddings. This formulation allows individual query terms to align with specific visual regions of the page, enabling a query about “capital expenditures” to align with the patches covering the relevant table row even if OCR of that region would be imperfect. Because patch embeddings can be precomputed offline, indexing a document corpus with ColPali is efficient and retrieval at query time requires only the MaxSim computation over pre-indexed patch embeddings.

On the DocVQA and ViDoRe benchmarks, ColPali substantially outperforms text-extraction-based baselines on pages with complex table layouts and embedded figures [Faysse et al., 2024], with the largest gains precisely on the content types that most challenge PDF parsing.

1.6.3. Recent Advances and Remaining Challenges

A limitation of the flat patch-aggregation approach in ColPali is that all patches of a page are indexed at a single level of granularity, regardless of their visual complexity or their relevance to typical queries. Recent work has explored multi-resolution indexing strategies that extend the patch-based paradigm to multiple spatial scales. Fang et al. [2025] propose a three-pass pyramid indexing framework that constructs separate visual representations at page, region, and patch levels, allowing the retrieval system to match queries at whichever visual granularity is most discriminative. On document-heavy benchmarks, this multi-scale approach improves recall on questions that require matching fine-grained visual elements such as specific table cells or column labels, where full-page patch aggregation tends to dilute the relevant signal across too many patches.

Despite these advances, visual retrieval methods face a set of challenges that remain partially unresolved. The resolution at which VLMs process document pages is constrained by the input size of the vision encoder, many financial tables use small fonts and dense layouts that are difficult to read at the resolutions typically used for training. Domain mismatch is also a concern, since most VLM-based retrieval models are trained on general document corpora and may not have seen sufficient examples of the standardized typography and table

structures used in SEC filings to develop discriminative representations for that domain. Finally, the computational cost of storing full patch embedding sequences for large document corpora is substantially higher than storing single-vector chunk embeddings, limiting the scalability of late-interaction visual retrieval. Whether domain-specific fine-tuning, higher-resolution encoders, or hybrid approaches combining text and visual signals can address these gaps is an open and active area of research.

1.7. Domain Adaptation for Financial Retrieval

Embedding models trained on general-domain corpora face a systematic mismatch between the distributional properties of their training data and the properties of financial documents and queries. Financial reports use a specialized vocabulary in which terms such as “EBITDA,” “net tangible assets,” and “diluted earnings per share” carry precise technical meanings that diverge from their general-language senses or have no general-language counterpart. Queries about financial metrics are typically short, formally phrased, and saturated with abbreviations, while the relevant passages are long, structurally repetitive, and express the same concepts using standardized financial language of SEC filings. This query-passage asymmetry is quite prominent in the financial domain, and it is compounded by the specificity of financial questions. A model that has not seen many examples of matching FY2022 queries to FY2022 evidence, rather than to structurally identical FY2021 or FY2023 passages, has no learned basis for that distinction.

The standard approach to bridging this domain gap is fine-tuning. Given domain-specific query-passage pairs, the embedding model is trained to score relevant pairs higher than irrelevant ones. For bi-encoder fine-tuning, the Multiple Negatives Ranking (MNR) loss is commonly used. It treats all other passages in the minibatch as implicit negatives for each query, creating a computationally efficient approximation to listwise ranking. For cross-encoder rerankers, binary cross-entropy (BCE) loss over explicit positive and sampled-negative pairs is preferred, as it calibrates a pointwise relevance score rather than requiring exhaustive listwise normalisation. Hard negatives, drawn from topically similar but answer-irrelevant passages rather than at random, play a critical role in shaping what either model learns, particularly in the financial domain where boilerplate language makes random negatives trivially easy to dismiss. Kim et al. [2025] demonstrate that fine-tuning BGE-M3 on financial query-passage pairs improves FinanceBench retrieval substantially relative to the zero-shot model, with the fine-tuned model showing improved ability to match financial terminology and temporal expressions. FinSage [Wang et al., 2025] applies direct preference optimization (DPO) on financial query-document pairs to train a reranker that learns fine-grained distinctions, including between passages from the same document that discuss the same topic in different

fiscal years. The specific loss formulations, negative sampling strategies, and hyperparameter choices for both the bi-encoder page scorer and the fine-tuned cross-encoder reranker used in this thesis are presented in detail in Chapter 3.

1.8. Summary

This chapter has traced the evolution of retrieval-augmented generation from its foundational contributions through the current state of the field, examined the information retrieval methods that form the technical core of RAG systems, and situated financial question answering as a domain that stresses these methods in ways that general-domain evaluations do not reveal. The survey of benchmarks established FinanceBench as the primary evaluation standard for this thesis while motivating its extension to address limitations in document diversity and coverage.

The treatment of hierarchical retrieval surveyed the landscape from general-purpose recursive indexing systems such as RAPTOR to domain-specific two-stage pipelines for financial documents. It treated chunking as an integral granularity decision within that hierarchy rather than a separate preprocessing step, identifying the interaction between chunk size and retrieval architecture as an understudied variable that this thesis addresses experimentally. Pages were identified as a natural and well-motivated intermediate retrieval unit in the context of long financial filings. Domain adaptation established the contrastive fine-tuning approaches that allow general embedding models to acquire the financial retrieval competence needed to distinguish temporal near-duplicates and match domain-specific terminology. The closing section on visual document retrieval introduced the ColPali architecture and recent multi-scale extensions as an emerging complementary paradigm whose applicability to financial documents is evaluated in the methodology that follows.

The literature reviewed in this chapter shows that financial RAG poses a retrieval problem that is more difficult than standard end-to-end evaluation. Long, repetitive, and table-heavy filings create challenges that motivate more fine-grained retrieval analysis and specialized retrieval architectures. The next chapters therefore turn to the datasets, indexing decisions, and evaluation setup used to study these challenges.

Chapter 2

Experimental Setup

2.1. Problem Formulation

Let $\mathcal{D} = \{d_1, \dots, d_{|\mathcal{D}|}\}$ be a corpus of financial filings, where each document $d \in \mathcal{D}$ consists of a set of pages $\mathcal{P}(d) = \{p_1^d, \dots, p_{n_d}^d\}$, each page being a sequence of extracted text. The full corpus is segmented into a chunk corpus

$$\mathcal{C} = \{(c, d, p) \mid c \in \text{chunks}(p), p \in \mathcal{P}(d), d \in \mathcal{D}\}, \quad (2.1.1)$$

where each chunk c carries document identifier d and page index p as metadata. Given a natural language question q , a retriever selects the top- k chunks

$$R_k(q) = \text{Top-}k_{(c,d,p) \in \mathcal{C}} s(q, c) = \{(c_i, d_i, p_i)\}_{i=1}^k, \quad (2.1.2)$$

and a generator produces an answer $\hat{y} = G_\theta(q, R_k(q))$. FinanceBench provides a gold filing $d^*(q)$ and a set of gold page indices $P^*(q)$ for each question q , enabling evaluation at both the document and page level.

2.2. Dataset

2.2.1. FinanceBench

Our primary evaluation dataset is the open-source subset of FinanceBench [Islam et al., 2023], comprising 150 question-answer pairs over 84 unique financial PDF documents. Each sample includes a natural-language question, a reference answer, a verbatim evidence string, and gold metadata identifying the source document and evidence page number. This page-level annotation is central to our evaluation as it provides the ground-truth labels for document recall, page recall, and oracle retrieval conditions.

The 150 questions span three question types with 50 questions each. Domain-relevant questions apply broadly across companies and require locating narrative disclosures; metrics-generated questions require extracting or computing specific financial indicators from tabular contexts; novel-generated questions test company-specific reasoning and are designed to require inference beyond simple extraction. The underlying documents span four filing types: annual 10-K reports account for $n = 112$ questions (74.7%), quarterly 10-Q reports for $n = 15$ (10.0%), earnings call transcripts for $n = 14$ (9.3%), and 8-K current reports for $n = 9$ (6.0%). The strong skew toward 10-K filings reflects both their prevalence in FinanceBench and their importance as the primary vehicle for annual financial disclosure.

The PDF documents are downloaded from the public FinanceBench release.¹ Following FinanceBench, page text is extracted using PyMuPDF from LangChain [Chase, 2022], preserving page boundaries as metadata. All page numbers in our evaluation correspond directly to the physical page indices in the PDF documents, validated against the gold annotations in FinanceBench.

2.2.1.1. Illustrative Examples by Question Type. To make the three question types concrete, Examples 1 through 3 present one representative example from each category, drawn from FinanceBench. Each example displays the natural-language question, the reference answer provided by the dataset, the verbatim gold-evidence string on which the answer is grounded, and the source filing metadata. These examples illustrate the qualitative differences in evidence location, answer type, and required reasoning that motivate the per-type analysis in Sections 4.2.5 and 4.3.5.

Example 1 | Domain-Relevant Question

Question: Among operating, investing, and financing activities, which brought in the most (or lost the least) cash flow for Nike in FY2023?

Expected answer: Cash flow from operations was the highest in FY2023.

Evidence: “NIKE, INC. CONSOLIDATED STATEMENTS OF CASH FLOWS. YEAR ENDED MAY 31 (Dollars in millions). 2023 2022 2021. Cash provided (used) by operations: Net income \$5,070 \$6,046 \$5,727. ... **Cash provided (used) by operations** **5,841 5,188 6,657.** ... **Cash provided (used) by investing activities** **564 (1,524) (3,800).** ... **Cash provided (used) by financing activities** **(7,447) (4,836) (1,459).** ...”

¹<https://github.com/patronus-ai/financebench>

Document Name: NIKE_2023_10K **Filing:** 10K **Gold Page:** p. 61

Example 2 | Metrics-Generated Question

Question: What is Amcor’s year end FY2020 net AR (in USD millions)? Address the question by adopting the perspective of a financial analyst who can only use the details shown within the balance sheet.

Expected answer: \$1616.00.

Evidence: “Amcor plc and Subsidiaries Consolidated Balance Sheet (in millions) As of June 30, 2020 2019 Assets Current assets: Cash and cash equivalents \$ 742.6 \$ 601.6; Trade receivables, net **1,615.9** 1,864.3; Inventories...”

Document Name: AMCOR_2020_10K **Filing:** 10K **Gold Page:** p. 49

Example 3 | Novel-Generated Question

Question: By how much did Pepsico increase its unsecured five year revolving credit agreement on May 26, 2023?

Expected answer: \$400,000,000 increase.

Evidence: “Effective May 26, 2023, PepsiCo terminated the **\$3,800,000,000 five year unsecured revolving credit agreement** (2022 Five Year Credit Agreement). ... On May 26, 2023, PepsiCo entered into a new **\$4,200,000,000 five year unsecured revolving credit agreement** (2023 Five Year Credit Agreement) among PepsiCo, as borrower, the lenders party thereto, and Citibank, N.A., as administrative agent. ...”

Document Name: PEPSICO_2023_8K **Filing:** 8K **Gold Page:** p. 1

2.2.2. Benchmark Extension: LoFIN and FinQA

FinanceBench’s 150-question open-source subset, while well-annotated, covers a limited range of companies, years, and document types. To study generalisation beyond this distribution, we construct an extended evaluation set starting from LoFIN, the benchmark introduced with HiREC [Choe et al., 2025], which is composed from FinQA [Chen et al., 2021] and SEC-QA [Lai et al., 2024].

FinQA [Chen et al., 2021] contains 8,281 question-answer pairs sourced from earnings reports and 10-K filings of S&P 500 companies, with an emphasis on numerical reasoning over tables and surrounding text. Each record includes a question, a structured numerical answer, and `pre_text/post_text` passages that bracket the relevant table (example in Figure B.1 in the Appendix). Because FinQA annotations reference source passages rather than explicit PDF page numbers, linking them to our corpus requires an alignment step described as a methodological contribution in Section 3.6.

In practice, only the FinQA portion of LoFIN could be included in the PDF-grounded benchmark used in this thesis. The available SEC-QA records did not contain gold evidence text or gold page text suitable for page alignment, so SEC-QA questions could not be mapped to reliable gold PDF pages.

2.3. PDF Acquisition Pipeline

A part of this work involved constructing a reliable PDF corpus from the EDGAR free online database. FinanceBench already provides PDFs, but the LoFIN/HiREC extension required recovering filings from EDGAR identifiers since the PDFs were not provided in the benchmark. The acquisition procedure first normalises company identifiers, resolves each ticker to its corresponding unique identifier (CIK), queries the EDGAR submissions metadata for the target accession number, and parses the filing index to select the primary filing document. Native PDF filings are downloaded directly while HTML filings are converted into page-preserving PDFs before indexing.

2.3.1. HTML-to-PDF Conversion

Many EDGAR filings, particularly older 10-K and 10-Q submissions, are submitted as HTML documents rather than native PDFs. For these filings, the HTML is rendered to PDF using a headless browser. Before rendering, the document is made self-contained by embedding linked images, stylesheets, and background resources. This avoids repeated network calls during rendering and improves reproducibility across machines.

Before rendering, the conversion procedure detects likely page boundaries in the live document structure. It uses explicit page containers, horizontal separators, page-number footers, roman numerals, appendix-style page markers, and related layout cues. Native page-break rules are then replaced with controlled page breaks, and oversized sections are scaled to reduce blank or truncated pages. The rendered PDF uses Letter-sized pages, printed backgrounds, minimal browser margins, and print-media emulation. Generated PDFs are validated through minimum file-size checks, PDF signature checks, and page counting when

a readable page structure is available. To ensure page boundaries were respected, a manual check was done on each retrieved PDF.

2.4. Document Indexing and Retrieval Infrastructure

2.4.1. Text Extraction and Page Representation

For each filing in the corpus, page text is extracted using PyMuPDF via LangChain’s `PyMuPDFLoader` interface, preserving page boundaries as metadata tuples (d, p) , where d is the document identifier and p is the 0-indexed page number. All retrieval methods share this same extractor.

PyMuPDF was chosen as the default extractor because it is fast, stable, and directly supported by LangChain’s PDF loading interface, and because FinanceBench [Islam et al., 2023] adopts it as its reference extractor. A comparison of extraction methods on the FinanceBench corpus is reported in Appendix D.1. Overall, the differences across extractors are small (at most 0.027 at page recall at $k = 5$ and 0.040 at page recall at $k = 10$), suggesting that extractor choice is not a primary driver of retrieval performance on this corpus.

2.4.2. Chunk Construction and Chunking Strategies

We build the chunk corpus $\mathcal{C}$ by splitting page text into overlapping spans. Each chunk is represented as a tuple (c, d, p) carrying the chunk text, the source document identifier, and the page index. The default configuration uses a recursive character splitter with a window of 1024 tokens and an overlap of 128 tokens, following Yepes et al. [2024]. To verify the suitability of this 1024-token configuration for the FinanceBench corpus and to ensure our baseline was optimized, we conducted a preliminary ablation over varying chunk sizes and strategies. The results and configurations of these variants are detailed in Appendix E.

Fixed-size naive chunking. Text is split at fixed character or token boundaries without regard to sentence or paragraph structure; included as a baseline to verify that recursive splitting outperforms naive splitting on financial text.

Recursive character splitting. The splitter attempts to split on a priority-ordered list of separators (paragraph breaks, newlines, periods, spaces), recursively falling back to shorter separators when a chunk exceeds the target size. This is the default configuration (1024-token window, 128-token overlap).

Semantic chunking. Consecutive sentences are grouped into chunks by detecting BGE-M3 embedding similarity drops between adjacent sentences, with a new chunk started whenever cosine similarity falls below a configurable threshold (default 0.6).

Parent-child chunking. Two levels of chunks are constructed: coarse parent chunks of 1024 tokens serving as generation context, and fine child chunks of 256 tokens indexed for retrieval and mapped back to their parent at generation time.

Page-aligned chunking. Chunk boundaries are constrained to fall within a single page, preventing chunks from spanning adjacent page boundaries and ensuring unambiguous page metadata for the page scorer pipeline.

Late chunking. A long document passage is encoded as a single sequence using BGE-M3, and span-level embeddings are extracted by mean-pooling token-level representations within each chunk’s token span [Günther et al., 2024]. Page assignment uses character-level offset tracking and the tokenizer’s `return_offsets_mapping` flag to recover the originating page for each span.

2.4.3. Vector Store: ChromaDB

All dense retrieval experiments use ChromaDB [Team, 2023] as the vector store. ChromaDB was selected over FAISS [Douze et al., 2025] primarily for its native support for structured metadata filtering. Each chunk in the index carries `doc_name` and `page_num` as metadata fields, and ChromaDB’s `where` filter allows oracle document and oracle page retrieval conditions to be implemented as first-class query constraints rather than as post-retrieval filtering heuristics.

A single ChromaDB collection is built once per (embedding model, chunk size, chunk overlap) configuration and reused across all retrieval experiments sharing that configuration. This ensures that comparisons across retrieval methods reflect genuine method differences rather than index construction variance.

2.5. Retrieval Systems

We evaluate eleven retrieval configurations, all operating over the same shared multi-document index unless otherwise noted. This shared-index design controls for implementation differences and ensures that metric differences across methods reflect genuine retrieval quality rather than index construction artefacts.

2.5.1. Baseline Retrievers

Dense retrieval maps both the query and each chunk to a shared vector space via a bi-encoder and ranks chunks by cosine similarity [Karpukhin et al., 2020]. We evaluate three dense backbones. BAAI/BGE-M3 [Chen et al., 2025] is a multilingual, multi-granularity model producing 1024-dimensional vectors with strong performance across retrieval benchmarks; it serves as the primary dense retriever and as the base model for the learned page

scorer. MPNet is a smaller general-purpose encoder producing 768-dimensional vectors, included as a lighter alternative to separate the effect of model capacity from pipeline design. `nickmuchi/finance-investopedia-embeddings` is a domain-specific model pre-trained on Investopedia text (768-dimensional vectors), included to test whether financial-domain vocabulary exposure alone, without retrieval-specific contrastive training, is sufficient to outperform general-purpose encoders on SEC filing retrieval.

For sparse retrieval, BM25 [Robertson and Zaragoza, 2009a] is implemented using the `rank-bm25` library with a finance-aware tokenizer that preserves currency symbols, percentage signs, and formatted numbers. SPLADE [Formal et al., 2021] expands query and document representations via a masked language model into sparse term-weighted vectors over the full vocabulary; we use the `naver/splade-cocondenser-ensembledistil` checkpoint, truncating inputs to 512 tokens and retaining the top 256 weighted terms per representation for index efficiency.

The hybrid baseline combines BM25 and BGE-M3 rankings using Reciprocal Rank Fusion (RRF) [Cormack et al., 2009]. For a chunk at rank position r , its RRF score is

$$\text{RRF}(r) = \frac{1}{k_{\text{rrf}} + r}, \quad (2.5.1)$$

with $k_{\text{rrf}} = 60$ following standard practice. Scores from both retrievers are summed and chunks are re-ranked by the combined score. We evaluate three mixing ratios (sparse-heavy 25/75, balanced 50/50, and dense-heavy 75/25) to establish the best hybrid configuration.

2.5.2. Query Reformulation

A deterministic preprocessing step replaces common financial abbreviations (e.g., EPS, EBITDA, FY) with their full-form equivalents and normalises fiscal year references to a canonical format (e.g., “FY2022” to “fiscal year 2022”), improving lexical overlap with filing text without requiring a generative model.

Hypothetical Document Embeddings (HyDE) [Gao et al., 2023a] generate a hypothetical passage that would answer the query and embed that passage rather than the query itself. We generate one hypothetical passage per query using Qwen-2.5-7B-Instruct at temperature 0.7 and embed it with BGE-M3. Multi-HyDE [George et al., 2025] reduces variance by sampling four independent hypothetical passages per query and averaging their BGE-M3 embeddings into a single query representation; the prompt template is reproduced in Appendix ??.

2.5.3. Reranking

We first implement a zero-shot cross-encoder via `BAAI/bge-reranker-v2-m3` serving as a reranker. The dense retriever first produces $N = 20$ candidate chunks, the cross-encoder

then assigns a scalar relevance score to each (query, chunk) pair via full cross-attention, and the top $k = 5$ chunks by reranker score are passed to the generator [Nogueira et al., 2019]. BGE-M3 + Multi-HyDE + ReRanker, which applies query augmentation before retrieval and reranks the resulting candidates, constitutes the strongest non-learned baseline.

We additionally fine-tune `BAAI/bge-reranker-v2-m3` on FinQA financial page-level data to adapt the reranker to the domain. The training procedure and loss function are described in Section 3.4; at inference time it operates identically to the zero-shot variant.

As an alternative second-stage model, we evaluate a late interaction reranker using ColBERT-style MaxSim scoring [Khattab and Zaharia, 2020]. The query is encoded as a sequence of token-level embeddings $\{\mathbf{q}_i\}$ and each candidate chunk as a sequence $\{\mathbf{d}_j\}$; relevance is computed as

$$s_{\text{MaxSim}}(q, c) = \sum_i \max_j \mathbf{q}_i^\top \mathbf{d}_j. \quad (2.5.2)$$

We use the `colbert-ir/colbertv2.0` checkpoint for this configuration.

2.6. Generation Setup

The generator model, prompt template, and decoding settings are held fixed across all retrieval methods to ensure that observed differences in answer quality reflect retrieval differences alone. We use Qwen-2.5-7B-Instruct [Team, 2025] as our primary generator, and use its 3B, 7B, and 14B variants to analyse the impact of scaling. The prompt follows a system-user template that presents the retrieved context and question, instructing the model to provide a concise, evidence-grounded answer in the style of a financial analyst. Greedy decoding is used with a maximum of 256 new tokens. The model is loaded in 8-bit quantization via `bitsandbytes` to reduce GPU memory consumption from approximately 16 GB to 9 GB, enabling experiments on a single consumer GPU. We additionally evaluate Llama-3.1-8B-Instruct [AI at Meta, 2024] as a baseline from a different generator family. The generator prompt template is given in full in Appendix A.1.

2.7. Evaluation Metrics

2.7.1. Retrieval Metrics

We begin by defining the core retrieval metrics used throughout this work, document and page recall. These metrics measure whether a system successfully retrieves the correct document and localizes the relevant evidence within it, reflecting the two primary stages at which retrieval can fail.

For each question q with gold filing $d^*(q)$ and gold pages $P^*(q)$, let the top- k retrieved chunks be $R_k(q) = \{(c_i, d_i, p_i)\}_{i=1}^k$. Document recall at k is the indicator that the gold filing appears among the retrieved chunks’ source documents,

$$\text{DocRec@}k(q) = \mathbb{I}[d^*(q) \in \{d_i\}_{i=1}^k]. \quad (2.7.1)$$

In the FinanceBench dataset, all questions can be answered by a single document. Page recall at k is the fraction of gold pages covered by chunks retrieved from the correct filing,

$$\text{PageRec@}k(q) = \frac{|P^*(q) \cap \{p_i \mid d_i = d^*(q)\}_{i=1}^k|}{|P^*(q)|}. \quad (2.7.2)$$

There can be more than one needed page to retrieve the full context required to answer a question. We report both metrics averaged over all 150 questions at $k = 5$. We prioritize recall because if the correct evidence is not retrieved, the generator cannot produce a correct answer. Missing even a single relevant page constitutes a failure mode that cannot be corrected downstream.

2.7.2. Context Quality Metrics

To measure how well the retrieved context captures the gold evidence independently of page-index accuracy, we report the maximum chunk-level BLEU [Papineni et al., 2002] and ROUGE-L [Lin, 2004] scores between any retrieved chunk and the gold evidence string $e^*(q)$,

$$\text{MaxBLEU@}k(q) = \max_{i=1}^k \text{BLEU}(c_i, e^*(q)), \quad \text{MaxROUGE-L@}k(q) = \max_{i=1}^k \text{ROUGE-L}(c_i, e^*(q)). \quad (2.7.3)$$

These metrics complement page-level recall since a method might achieve high page recall yet still produce low context quality if chunk boundaries do not align well with the evidence text. Conversely, context quality can be high even when page recall is imperfect, if the retrieved chunks happen to overlap the gold evidence despite originating from a different page index. This situation is noted in Section 4.2.5, where a retrieved chunk contains the correct information but comes from a different page than the annotation.

2.7.3. Answer Quality Metrics

For downstream generation quality, we report ROUGE-L between the generated answer $\hat{y}$ and the reference answer y^* . We additionally report the numeric match accuracy. Given reference answer y^* and generated answer $\hat{y}$, we extract the sets of numeric tokens N^* and $\hat{N}$ using a regular expression that strips commas and currency symbols, and count the question as correct if there exists a pair $(n^*, \hat{n}) \in N^* \times \hat{N}$ satisfying

$$|\hat{n} - n^*| \leq \varepsilon_{\text{abs}} \quad \text{or} \quad \frac{|\hat{n} - n^*|}{n^*} \leq \varepsilon_{\text{rel}}, \quad (2.7.4)$$

with $\varepsilon_{\text{abs}} = \varepsilon_{\text{rel}} = 0.03$, following Islam et al. [2023]. Numeric match is reported on all questions but since metrics-generated questions are always numeric, we provide a separate numeric match result on this subset of questions to remove noise from potential noisy results from the other question types.

2.7.4. LLM-Based and Human Evaluation

LLM-as-a-judge frameworks is not used as metrics in this thesis, this is motivated by several considerations. LLM-based evaluation is inherently stochastic and sensitive to prompt design and model choice, making results difficult to replicate. Prior work has identified systematic biases in LLM judges, including positional bias, verbosity bias, and sensitivity to answer phrasing [Zheng et al., 2023]. Computational constraints also make large-scale evaluation with high-quality judge models impractical. Instead of relying on automated LLM judges, the primary generative results reported rely on a combination of deterministic evaluation metrics (such as the numeric match) and manual human evaluation.

Because the evaluated FinanceBench subset consists of a manageable 150 questions, conducting a comprehensive manual evaluation is feasible. To maintain a strict standard of quality, the manual evaluation follows binary grading (correct or incorrect). Partial credit was not awarded, so if a generated answer was only partially correct, it was marked as incorrect. Furthermore, the evaluation prioritized the validity of the reasoning process over the final output alone. If a model generated the correct final answer but relied on flawed reasoning, or arrived at the right answer by accident, it was strictly marked as incorrect. This conservative approach ensures that the reported human accuracy metrics reflect genuine comprehension and evidence grounding rather than lucky guesses or hallucinated calculations.

2.8. Computational Infrastructure

Each job was allocated one NVIDIA RTX 3090 GPU (24 GB VRAM), 8 CPU cores, and between 32 GB and 64 GB of RAM depending on the experiment. Generator models (Qwen-2.5-7B-Instruct and Llama-3.2-3B-Instruct) are loaded in 8-bit precision using `bitsandbytes` [Dettmers et al., 2022], which reduces memory consumption by approximately half relative to 16-bit loading. Embedding models (BGE-M3, BGE-base, and the Investopedia encoder) are loaded in full 16-bit precision.

Chapter 3

Methodology

3.1. Overview

This chapter presents the methodological contributions of this thesis. The first is an oracle-based evaluation framework that decomposes retrieval failures into document-level and page-level components, providing empirical upper bounds at each granularity and a diagnostic framework for attributing performance deficits to their constituent sources. The second is a hierarchical page-then-chunk retrieval pipeline which leverages a learned page scorer by domain fine-tuning a bi-encoder based on BAAI/BGE-M3 that treats pages as a retrieval unit between documents and chunks. The third is a domain fine-tuned cross-encoder reranker based on BGE-reranker-v2-m3, trained on FinQA data and evaluated zero-shot on FinanceBench, enabling a controlled comparison between zero-shot and domain-adapted reranking. The fourth evaluates ColPali visual page scoring as an alternative Stage 1 ranker within the same pipeline architecture, allowing a direct comparison between text-based and image-based page representations on financial documents. The fifth describes the FinQA alignment pipeline used to extend FinanceBench with verified page-level supervision from EDGAR filings, including the fuzzy-matching procedure and the exclusion criteria applied to unverifiable examples.

3.2. Oracle-Based Evaluation Framework

A RAG pipeline can fail at two distinct stages. The retriever may fail to surface the relevant evidence, or the generator may fail to answer correctly even when the relevant evidence is present. Measuring end-to-end accuracy alone conflates these two sources and makes it impossible to identify the primary bottleneck. To separate them, we construct oracle retrieval conditions that progressively restrict the candidate pool to the gold evidence, yielding empirical upper bounds on what retrieval improvements alone could achieve.

3.2.1. Oracle Retrieval Conditions

We denote the full chunk corpus by $\mathcal{C}$, where each chunk is a tuple (c, d, p) of text, document identifier, and page number, as defined in Section 2.1. For each question q , FinanceBench provides a gold document d^* and a set of gold pages P^* . We define three candidate sets by successively narrowing the search space.

First, standard retrieval, where the retriever searches the full corpus $\mathcal{C}$. This is the baseline setting.

Second. Oracle document retrieval, where the candidate set is restricted to chunks originating from the gold filing,

$$\mathcal{C}^{\text{doc}}(d^*) = \{ (c, d, p) \in \mathcal{C} \mid d = d^* \}. \quad (3.2.1)$$

This eliminates cross-document retrieval errors entirely. Any remaining failure is attributable to within-document page or chunk selection.

Finally, Oracle page retrieval, where the candidate set is further restricted to chunks from the gold pages,

$$\mathcal{C}^{\text{page}}(d^*, P^*) = \{ (c, d, p) \in \mathcal{C} \mid d = d^*, p \in P^* \}. \quad (3.2.2)$$

The gold evidence is guaranteed to appear in the candidate pool, the only remaining source of retrieval failure is intra-page chunk quality and ranking.

The three candidate sets satisfy strict containment,

$$\mathcal{C}^{\text{page}}(d^*, P^*) \subset \mathcal{C}^{\text{doc}}(d^*) \subset \mathcal{C}. \quad (3.2.3)$$

Both oracle conditions are implemented via ChromaDB’s metadata filtering API, which restricts nearest-neighbour search to chunks whose `doc_name` matches d^* and, under oracle page conditions, whose `page_num` belongs to P^* . This ensures that the only variable across conditions is the candidate set, the embedding model and similarity function remain identical.

3.2.2. Oracle Conditions as Empirical Upper Bounds

Because each oracle candidate set is a subset of the previous one that is guaranteed to contain the gold evidence, restricting the search space reduces noise and therefore improves retrieval quality. Let $\mathcal{R}_k(q; \mathcal{X})$ denote the top- k chunks retrieved when searching candidate set $\mathcal{X}$, and let $G_\theta(q, \mathcal{R}_k)$ denote the generator output given those chunks. For any robust¹ retrieval metric R-Score and any generation metric G-Score, the containment in

¹Note: Here robust implies retrieval

Equation (3.2.3) implies, in expectation over the evaluation set,

$$\text{R-Score}(\mathcal{R}_k(q; \mathcal{C})) \leq \text{R-Score}(\mathcal{R}_k(q; \mathcal{C}^{\text{doc}})) \leq \text{R-Score}(\mathcal{R}_k(q; \mathcal{C}^{\text{page}})), \quad (3.2.4)$$

$$\text{G-Score}(G_\theta(q, \mathcal{R}_k(q; \mathcal{C}))) \leq \text{G-Score}(G_\theta(q, \mathcal{R}_k(q; \mathcal{C}^{\text{doc}}))) \leq \text{G-Score}(G_\theta(q, \mathcal{R}_k(q; \mathcal{C}^{\text{page}}))). \quad (3.2.5)$$

The retrieval bound in Equation (3.2.4) holds because removing non-gold documents and pages eliminates distractors that would otherwise displace gold chunks from the top- k list. The generation bound in Equation (3.2.5) follows directly: since the generator G_θ is fixed across conditions, improvements in retrieved evidence quality propagate to downstream answer quality.

Equation (3.2.5) provides the diagnostic core of this framework. Any gap between end-to-end performance under standard retrieval and under oracle page retrieval is attributable purely to retrieval failure, since the generator is identical across conditions and differs only in the evidence it receives. These oracle conditions should be read as empirical diagnostic bounds rather than mathematical guarantees for every possible model. They restrict the candidate set but do not directly optimize the final ranking, so a learned reranker or a domain-tuned retriever can exceed an oracle-document score by using ranking signals that the fixed oracle retriever does not use. This is why the experiments interpret oracle results as reference points for localizing failure, not as absolute ceilings on all architectures.

3.2.3. Retrieval Gap Decomposition

The three-level containment motivates a two-component decomposition of the overall performance deficit.

Within-document retrieval gap.

$$\Delta_{\text{page}} = \text{PageRec}@k(\mathcal{C}^{\text{doc}}) - \text{PageRec}@k(\mathcal{C}). \quad (3.2.6)$$

This measures how much page recall is lost to within-document retrieval errors once document discovery is held perfect. A large Δ_{page} indicates that the system correctly identifies the gold filing but fails to rank the answer-bearing pages highly within it.

Generation ceiling gap.

$$\Delta_{\text{gen}} = \text{NumMatch}(\mathcal{C}^{\text{page}}) - \text{NumMatch}(\mathcal{C}^{\text{doc}}). \quad (3.2.7)$$

This measures the gain in downstream answer accuracy that would result from perfect page localisation, holding document discovery constant. A large Δ_{gen} indicates that precise page retrieval is the binding constraint on end-to-end accuracy.

Together, Δ_{page} and Δ_{gen} attribute the total performance deficit to its two constituent sources and provide a principled basis for prioritizing system improvements, if Δ_{page} dominates, the highest return lies in better within-document page retrieval, if Δ_{gen} dominates, finer-grained evidence localization at the chunk level is the priority.

3.3. Hierarchical Page-then-Chunk Retrieval

3.3.1. Motivation and Architecture

Standard chunk retrieval searches $\mathcal{C}$ directly, competing against all chunks in the corpus simultaneously. For long financial filings with repeated templates, dense tables, and structurally similar sections across years, this means that the retriever must simultaneously distinguish the correct filing from thousands of similar passages and locate the correct page within hundreds of pages in that filing. These two tasks have different natural difficulty levels, and conflating them in a single retrieval step means that the method optimized for document-level discrimination may not be optimal for page-level localization within the correct document.

The proposed hierarchical pipeline separates these tasks explicitly. In Stage 1, a learned page scorer ranks all pages in the corpus and selects the top- P pages. In Stage 2, standard dense chunk retrieval is applied, but restricted to chunks from those P pages. This design reduces the effective search space for chunk retrieval from $|\mathcal{C}|$ to approximately $P \cdot \bar{n}$ chunks, where $\bar{n}$ is the average number of chunks per page. With $P = 20$, this represents a reduction of roughly two orders of magnitude for a corpus of the FinanceBench scale. The filtering also concentrates chunk retrieval on the most semantically relevant regions of the corpus, reducing noise from unrelated sections while preserving fine-grained selection within the selected pages.

3.3.2. The Learned Page Scorer

For each filing d and page p , we extract the normalized page text $\tilde{t}_{d,p} = \mathcal{N}(t_{d,p})$ as described in Section 2.4 and encode it with a fine-tuned bi-encoder E_{θ}^{page} . The page relevance score for query q is the cosine similarity between the query embedding and the page embedding,

$$s_{\text{page}}(q, d, p) = \cos(E_{\theta}^{\text{page}}(q), E_{\theta}^{\text{page}}(\tilde{t}_{d,p})). \quad (3.3.1)$$

The same encoder is used for both the query and the page text; the model learns to represent queries and pages in a shared space where relevant pairs are close. The top- P pages selected by Stage 1 are

$$\mathcal{P}_P(q) = \text{Top-}P_{(d,p) \in \mathcal{P}} s_{\text{page}}(q, d, p), \quad (3.3.2)$$

and the filtered chunk corpus is

$$\mathcal{C}_{\text{filt}}(q) = \{(c, d, p) \in \mathcal{C} \mid (d, p) \in \mathcal{P}_P(q)\}. \quad (3.3.3)$$

Stage 2 then runs the standard dense chunk retriever over $\mathcal{C}_{\text{filt}}(q)$ and returns the top- k chunks for generation. The full routine is given in Algorithm 1.

Algorithm 1 Page-then-chunk retrieval with the learned page scorer.

Require: Query q ; fine-tuned page encoder E_{θ}^{page} ; chunk encoder E^{chunk} ; page index $\mathcal{I}_{\text{page}}$ storing $(d, p, \mathbf{v}_{d,p}^{\text{page}})$ with $\mathbf{v}_{d,p}^{\text{page}} = E_{\theta}^{\text{page}}(\tilde{t}_{d,p})$ precomputed offline; chunk index $\mathcal{I}_{\text{chunk}}$ storing $(c, \mathbf{v}_c^{\text{chunk}}, d_c, p_c)$ with $\mathbf{v}_c^{\text{chunk}} = E^{\text{chunk}}(c)$ precomputed offline; number of pages P ; number of chunks k .

1: **Stage 1 (page retrieval):**

2: $\mathbf{e}_q^{\text{page}} \leftarrow E_{\theta}^{\text{page}}(q)$

3: $\mathcal{P}_P(q) \leftarrow \text{SearchCos}(\mathcal{I}_{\text{page}}, \mathbf{e}_q^{\text{page}}, P)$ {nearest-neighbour search over stored $\mathbf{v}_{d,p}^{\text{page}}$ }

4: **Stage 2 (chunk retrieval on the filtered search space):**

5: $\mathbf{e}_q^{\text{chunk}} \leftarrow E^{\text{chunk}}(q)$

6: $\mathcal{C}_{\text{filt}} \leftarrow \{c \in \mathcal{I}_{\text{chunk}} \mid (d_c, p_c) \in \mathcal{P}_P(q)\}$

7: $R_k(q) \leftarrow \text{SearchCos}(\mathcal{C}_{\text{filt}}, \mathbf{e}_q^{\text{chunk}}, k)$ {search over stored $\mathbf{v}_c^{\text{chunk}}$ in the filtered pool}

8: **return** $R_k(q)$

The ablation in Appendix D shows that $P = 20$ achieves the best balance between recall and noise reduction, with little gain from larger values of P .

3.3.3. Training Procedure

The page scorer is trained by fine-tuning BAAI/BGE-M3 as a bi-encoder for page-level relevance on financial filings. The model is initialized from the pretrained BGE-M3 checkpoint and optimized using the Multiple Negatives Ranking (MNR) loss [Henderson et al., 2017]. For a minibatch of B positive (query, page) pairs $\{(q_i, \tilde{t}_{d_i^*, p_i^+})\}_{i=1}^B$, the loss is

$$\mathcal{L}_{\text{MNR}} = -\frac{1}{B} \sum_{i=1}^B \left[s_{\text{page}}(q_i, d_i^*, p_i^+) - \log \sum_{j=1}^B \exp s_{\text{page}}(q_i, d_j^*, p_j^+) \right]. \quad (3.3.4)$$

For each query q_i , its paired gold page is the positive target and all other gold pages in the minibatch serve as in-batch negatives. Because questions in the same minibatch come from different companies and fiscal years, the negatives are *semantically hard*: they describe the same types of financial information (income statements, balance sheets) but from different entities, forcing the model to learn company- and year-specific numerical grounding rather than relying on generic financial vocabulary.

Training data is constructed from the 150 FinanceBench annotations. For each question, we pair it with every annotated gold page in the gold filing, yielding 170 positive pairs (some questions have multiple gold pages). To prevent data leakage, questions are partitioned by source document so that no document appears in both training and evaluation folds. We perform 5-fold cross-validation, training on 80% of documents and evaluating on the remaining 20% in each fold.

Optimization uses AdamW with a learning rate of 2×10^{-5} , linear warmup over the first 10% of training steps, and cosine decay thereafter. Each fold is trained for 15 epochs with a batch size of 16, maximum sequence length of 512 tokens, and early stopping with patience of 3 evaluations (evaluated every 3 epochs). Training converges in approximately 20 minutes per fold on a single NVIDIA RTX 3090.

3.3.4. Key Design Choices

Several design choices distinguish the learned page scorer from prior hierarchical retrieval methods.

The choice of *pages* as the retrieval unit, rather than arbitrary passages or sections, is motivated by their unambiguous boundaries, their correspondence to coherent presentation units in financial filings, and their direct annotation in FinanceBench, which provides supervision unavailable for sub-page passage boundaries.

The choice to *fine-tune a separate page encoder* rather than reusing the zero-shot chunk-level retriever explicitly teaches the model the asymmetry between short analytical queries and long, structured financial page text.

The use of *document-level cross-validation splits* prevents the model from memorizing filing-specific patterns and provides a cleaner estimate of generalization. Using question-level splits instead would allow questions about the same filing to appear in both training and test folds, inflating reported performance.

Finally, using *in-batch negatives* from the gold pages of other questions provides hard negatives without requiring explicit offline negative mining, which would require a separate retrieval pass and introduces risk of false negatives (pages that are actually relevant but not annotated as gold).

3.4. Fine-Tuned Cross-Encoder Reranker

3.4.1. Motivation

The zero-shot cross-encoder reranker (Section 2.5.3) applies a general-purpose relevance model to financial passages without any domain-specific adaptation. While this is a strong

baseline, financial filings have a distinctive vocabulary and structure—standardised line-item labels, tabular numerical evidence, SEC section headers—that a general model may not score optimally. To quantify how much domain fine-tuning helps, we train a second cross-encoder on FinQA financial page data and evaluate it on the held-out FinanceBench benchmark.

3.4.2. Architecture

The fine-tuned reranker is initialized from BAAI/bge-reranker-v2-m3, a multilingual cross-encoder pre-trained for passage reranking. The model takes as input the concatenation $[q; [\text{SEP}]; d]$ of a query q and a passage d , passes the sequence through the transformer stack, and applies a linear projection to the $[\text{CLS}]$ representation to produce a scalar relevance score

$$s_{\theta}(q, d) = \mathbf{w}^{\top} \text{CLS}(E_{\theta}([q; [\text{SEP}]; d])). \quad (3.4.1)$$

The maximum input length is capped at 512 tokens, which accommodates both the short financial query and a substantial portion of the page text.

3.4.3. Training Data

Training data is constructed from the FinQA training set [Chen et al., 2021], which contains 6,251 financial question-answer pairs annotated against pages from SEC annual report filings. For each FinQA training sample, we build the gold text by concatenating the pre-text, formatted table, and post-text fields (Appendix B.1) corresponding to the annotated gold evidence:

$$t_{\text{page}} = \text{pre_text} \parallel \text{flatten}(\text{table}) \parallel \text{post_text}. \quad (3.4.2)$$

This produces one positive pair $(q, t_{\text{page}}, 1)$ per sample.

For hard negatives, we sample $m = 3$ pages from other questions’ gold pages, specifically from questions whose source document differs from the current sample. This creates inter-document negatives, financially plausible pages (income statements, balance sheets from other companies or years) that are wrong answers to the current question. We deliberately avoid intra-document negatives, passages retrieved from within the same page as the gold evidence, because our retrieval task is inter-page (ranking different pages against each other), and training on within-page negatives would teach the wrong discrimination skill. Each training batch therefore contains one positive and three negative pairs per question, giving roughly $6,251 \times 4 = 25,004$ training examples.

3.4.4. Loss Function

The reranker is trained with binary cross-entropy loss applied to the sigmoid-transformed scalar score:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log \sigma(s_{\theta}(q_i, d_i)) + (1 - y_i) \log(1 - \sigma(s_{\theta}(q_i, d_i))) \right], \quad (3.4.3)$$

where $y_i \in \{0, 1\}$ is the binary relevance label and σ is the sigmoid function. This loss directly optimises the model to assign high scores to positive (query, page) pairs and low scores to negative pairs, without requiring the listwise normalisation required by MNR loss. Binary cross-entropy is appropriate here because the negatives are sampled rather than exhaustive, so the model learns a calibrated relevance signal rather than a relative rank within a fixed candidate list.

3.4.5. Training Procedure

Training uses the AdamW optimiser with a learning rate of 2×10^{-5} , a linear warmup over the first 10% of steps, and no additional decay schedule. We train for 3 epochs with a batch size of 32 pairs; at this batch size and a sequence length of 512 tokens under FP16 mixed precision, memory consumption is within the 46 GB VRAM budget of the RALI L40S GPU. The best checkpoint is selected by MAP on the FinQA development set, evaluated using the `CERerankingEvaluator` from `sentence-transformers`. Training and evaluation together require approximately four hours on a single L40S.

Table 3.1 summarises the hyperparameters used for both the bi-encoder page scorer (Section 3.3.3) and the cross-encoder reranker, enabling direct comparison of the two fine-tuning regimes.

At inference time, the fine-tuned cross-encoder reranker operates identically to its zero-shot counterpart. BGE-M3 dense retrieval first produces $N = 20$ candidate chunks, the reranker scores each (query, chunk) pair in batches of 64, and the top-5 chunks by reranker score are returned to the generator.

3.5. Visual Page Scoring with ColPali

As discussed in Section 1.6, visual retrieval models offer an alternative approach to Stage 1 page scoring that bypasses text extraction entirely. Rather than representing pages as sequences of extracted tokens, a vision-language model encodes each page as a sequence of patch embeddings derived from the rendered page image. This approach preserves the spatial layout of tables and multi-column structures that text extraction frequently distorts,

Table 3.1. Hyperparameters for the bi-encoder page scorer and the fine-tuned cross-encoder reranker.

Hyperparameter	Bi-Encoder (Page Scorer)	Cross-Encoder (Reranker)
Base model	BAAI/bge-m3	BAAI/bge-reranker-v2-m3
Training data	FinanceBench (5-fold CV)	FinQA train (6,251 samples)
Loss function	MNR (in-batch negatives)	BCE (sampled negatives)
Negatives per query	In-batch (15 per step)	3 inter-document
Learning rate	2×10^{-5}	2×10^{-5}
Batch size	16	32
Epochs	15	3
Warmup	10% of steps	10% of steps
Max sequence length	512 tokens	512 tokens
Precision	FP32	FP16 (AMP)
GPU	RTX 3090	L40S

and may capture visual cues such as font weight and column alignment that are semantically significant but invisible to text-based encoders.

To evaluate whether visual page scoring can serve as a viable or complementary alternative, we replace the fine-tuned text-based bi-encoder in Stage 1 with ColPali [Faysse et al., 2024]. Under this variant, each page is represented offline as a set of patch embeddings $\{\mathbf{e}_j^p\}$ produced by the PaLIGemma vision encoder. At query time, the relevance score for page p is computed using the ColBERT-style MaxSim operator,

$$s_{\text{ColPali}}(q, p) = \sum_i \max_j \mathbf{q}_i^\top \mathbf{e}_j^p, \quad (3.5.1)$$

where $\{\mathbf{q}_i\}$ are the token-level query embeddings. The top- P pages are selected by this visual score, and Stage 2 chunk retrieval proceeds identically over the filtered page set using the same BGE-M3 chunk encoder. This design isolates the contribution of the Stage 1 scoring function: all downstream behaviour is held constant, and only the page ranking criterion changes between the text-based and visual variants.

One practical difference between the two variants is indexing cost. Text-based page embeddings are single 1024-dimensional vectors per page, while ColPali stores a full sequence of patch embeddings per page (typically several hundred vectors), increasing the Stage 1 index footprint by roughly two orders of magnitude and requiring a MaxSim scan at retrieval time rather than a standard nearest-neighbour lookup. For the FinanceBench corpus scale this remains tractable, but the difference is relevant when considering deployment at scale.

3.6. Benchmark Extension: FinQA Alignment Pipeline

The benchmark extension used in this thesis is FinQA-only. HiREC releases the broader LOFin benchmark [Choe et al., 2025], but the local SEC-QA records did not contain gold evidence text or gold page text that could be mapped reliably to a PDF page. SEC-QA was therefore not added to the page-level benchmark, because downloadable PDFs alone are not enough for PageRec@ k evaluation since every question needs a verified gold page.

Each LoFIN FinQA example contains a structured identifier encoding the company, fiscal year, and source page. This identifier is matched to the corresponding original FinQA record, and the surrounding evidence text is used as an anchor for page recovery. For each candidate filing, the page text is extracted and compared with the evidence anchor using token-set fuzzy matching. The highest-scoring page is accepted only when its normalised similarity is at least $\tau = 0.70$, otherwise the example is excluded from page-level supervision.

The alignment processes 530 LoFIN FinQA rows, matched all 530 identifiers to their original FinQA records, and produced PDF-verified page evidence for 503 rows (94.9%). The remaining 27 rows were not used as verified page-supervision examples: 13 fell below the fuzzy-match threshold and 14 had no matching PDF in the local corpus. Main FinanceBench indexing still uses PyMuPDF/PyMuPDFLoader as described in Section 2.4; the FinQA alignment step uses a separate lightweight page-text extraction procedure because it only needs text for fuzzy evidence matching.

Chapter 4

Results and Analysis

This chapter presents and analyses all experimental results in three main parts. Section 4.1 establishes oracle-based empirical upper bounds that serve as reference points throughout. Sections 4.2 and 4.3 evaluate retrieval and generation respectively, each following the same three-level structure: aggregate performance, stratified breakdowns by question and document type, and analysis. Section 4.4 reports generalisation to the benchmark extension.

4.1. Oracle Analysis

Table 4.1 establishes empirical upper bounds under two idealised retrieval conditions evaluated at $k = 5$ using BGE-M3. Under oracle document retrieval, where the search space is restricted to the gold filing, page recall reaches 0.60 and ROUGE-L reaches 0.42. Under oracle page retrieval, where the search space is further restricted to the annotated gold pages, page recall reaches 1.00 by construction and ROUGE-L rises to 0.59. The corresponding generation results, reported in Table 4.5 using Qwen-2.5-7B-Instruct, show numeric match improving from 0.44 under oracle document to 0.70 under oracle page on the 50 metrics-generated questions.

Table 4.1. Oracle retrieval results at $k = 5$ using BGE-M3. BLEU and ROUGE-L measure context quality (retrieved chunks vs. gold evidence string). Numeric match is drawn from the downstream generation evaluation in Table 4.5 (Qwen-2.5-7B-Instruct, metrics-generated questions only).

Setting	DocRec@5	PageRec@5	BLEU@5	ROUGE-L@5	Num. Match
Oracle document	1.00	0.60	0.25	0.42	0.44
Oracle page	1.00	1.00	0.40	0.59	0.70

Two diagnostic gaps follow from these bounds. The *within-document retrieval gap* measures headroom available after perfect document discovery: any system’s PageRec@5 below 0.60 is attributable entirely to within-document page selection failure, not to document confusion. The *generation ceiling gap* is the difference in numeric match between oracle document (0.44) and oracle page (0.70), a difference of 0.26 points that quantifies the maximum generation improvement achievable purely from better page localisation, holding the generator constant. The fact that oracle page ROUGE-L (0.59) substantially exceeds oracle document ROUGE-L (0.42) confirms that page localisation quality, not document discovery, is the binding bottleneck on context quality. Notably, oracle page BLEU (0.40) also exceeds oracle document ROUGE-L (0.42) entirely on the strength of evidence concentration: restricting retrieval to gold pages eliminates off-target chunks and forces the top- k to be drawn from a small, evidence-dense pool. Both bounds recur as reference points throughout the chapter.

4.2. Retrieval Evaluation

This section evaluates all retrieval configurations at three levels of granularity. Aggregated results over the full 150-question benchmark establish the overall ranking of methods and identify the dominant failure modes. Stratified breakdowns by question type and document type then decompose these aggregate figures to show where each method succeeds and where it fails, and why. A budget analysis examines how recall evolves as the retrieval budget k increases, separating structural retrieval failures from cases where a larger candidate set would suffice. The section closes with a qualitative analysis that grounds the quantitative patterns in concrete examples drawn from question types.

4.2.1. Aggregated Retrieval Performance

Table 4.2 reports DocRec@5, PageRec@5, BLEU@5, and ROUGE-L@5 across all configurations evaluated on the full 150 FinanceBench questions at $k = 5$.

The most consistent finding across all configurations is that page recall lags document recall by a wide margin. Dense BGE-M3 achieves DocRec@5 of 0.94 but only PageRec@5 of 0.35, a gap of 0.59 points that is representative of the full table. Even the strongest non-proposed baseline, BGE-M3 + Multi-HyDE + ReRanker at PageRec@5 of 0.48, retains a 0.12-point gap to the oracle document bound of 0.60. This persistent gap confirms that the primary bottleneck is within-document localisation, not document discovery, and motivates the page-level supervision introduced in Section 3.3.

Lexical methods suffer severely on this corpus. BM25 reaches DocRec@5 of only 0.55 and PageRec@5 of 0.11, reflecting a fundamental mismatch between conversational query

Table 4.2. Aggregated retrieval and context-quality performance over 150 FinanceBench questions at $k = 5$. BLEU and ROUGE-L measure context quality (retrieved chunks vs. gold evidence string). Bold marks the best non-oracle result per column.

Method	DocRec@5	PageRec@5	BLEU@5	ROUGE-L@5
<i>Sparse</i>				
BM25	0.55	0.11	0.07	0.15
SPLADE	0.81	0.31	0.24	0.36
<i>Dense</i>				
MPNet	0.77	0.33	0.24	0.34
Finance-Investopedia	0.52	0.10	0.09	0.19
Dense BGE-M3	0.94	0.35	0.25	0.35
BGE-M3 + RR	0.95	0.44	0.29	0.40
<i>Hybrid</i>				
RRF 25/75 (sparse-heavy)	0.62	0.11	0.07	0.15
RRF 50/50	0.87	0.27	0.19	0.28
RRF 75/25 (dense-heavy)	0.93	0.32	0.22	0.32
<i>Hierarchical and query reformulation</i>				
Parent-Child	0.91	0.32	0.23	0.34
Query Expansion	0.93	0.38	0.26	0.37
BGE-M3 + HyDE	0.91	0.40	0.26	0.37
BGE-M3 + Multi-HyDE	0.89	0.44	0.27	0.39
BGE-M3 + Multi-HyDE + RR	0.93	0.48	0.32	0.43
<i>VLM</i>				
ColPali (standalone visual)	0.21	0.01	0.00	0.02
ColPali rerank (BGE-M3 + ColPali RR)	0.93	0.29	0.19	0.28
<i>Proposed</i>				
Page-then-chunk (in-domain)	0.95	0.55	0.33	0.46
BGE-M3 + Multi-HyDE + FT-RR	0.95	0.56	0.33	0.44
Oracle document	1.00	0.60	0.25	0.42
Oracle page	1.00	1.00	0.40	0.59

phrasing and the formalised vocabulary of SEC filings, where concepts are expressed through standardised line-item labels and regulated section headers that rarely appear verbatim in analyst-style questions. Sparse-heavy hybrid fusion (RRF 25/75) does not recover this deficit: incorporating the BM25 signal at 25% dense weight introduces sufficient rank noise to reduce DocRec@5 to 0.62, below even SPLADE alone (0.81). Increasing the dense weight (RRF 75/25) recovers most of the dense performance but never surpasses it, confirming that BM25 contributes no complementary signal on this corpus that compensates for the noise it introduces. SPLADE partially recovers from the sparse shortfall through learned vocabulary expansion (DocRec@5 of 0.81, PageRec@5 of 0.31) but remains well below the dense BGE-M3 baseline because its expansion vocabulary is still grounded in term co-occurrence patterns that do not capture the semantic relationships between query intent and filing structure.

Finance-Investopedia is the single worst dense model, with DocRec@5 of 0.52 and PageRec@5 of 0.10, underperforming BM25 on document recall and matching it on page recall. This model was pre-trained on Investopedia articles and financial news, a domain that shares surface-level financial terminology with SEC filings but differs structurally: news text is discursive and entity-centric, whereas SEC filings are tabular, numerically dense, and constrained by regulatory disclosure requirements. The embedding space learned from news content maps query embeddings to a representation region that is poorly aligned with the formal filing passages, making this a clear case of negative domain transfer.

MPNet achieves DocRec@5 of 0.77 and PageRec@5 of 0.33, approaching SPLADE on document recall despite being a general-purpose dense model with no financial pretraining. Its BLEU@5 (0.24) and ROUGE-L@5 (0.34) are comparable to SPLADE, suggesting that a well-trained general sentence encoder already captures enough semantic proximity to outperform lexical methods on this task. The gap relative to BGE-M3 (DocRec@5 of 0.94) is attributable to BGE-M3’s larger parameter count, its multilingual training over a far broader corpus, and its fine-tuning regime that includes hard-negative mining on retrieval datasets.

Query reformulation provides the largest single gains among non-fine-tuned approaches. HyDE (0.40 PageRec@5) and Multi-HyDE (0.44) outperform the dense baseline (0.35) by a substantial margin, demonstrating that generating a hypothetical financial passage before retrieval effectively bridges the vocabulary gap between short analytical queries and the dense technical language of evidence pages. Multi-HyDE’s gain over single HyDE reflects the benefit of variance reduction: averaging over multiple independently generated hypothetical documents smooths out idiosyncratic phrasing from any individual generation, yielding a more stable query representation. Adding the zero-shot cross-encoder reranker to Multi-HyDE raises PageRec@5 to 0.48, a further 0.04-point gain that arises because the reranker is sensitive to fine-grained numerical and structural cues that cosine similarity between aggregate embeddings cannot capture.

ColPali in standalone visual retrieval mode (PageRec@5 of 0.01, DocRec@5 of 0.21) is essentially non-functional on this benchmark. The failure reflects the domain gap between ColPali’s training distribution, which consists of diverse natural document types, and the dense numerical layout of financial statement pages. When ColPali is used as a reranker over BGE-M3 candidates (“ColPali rerank”), meaning BGE-M3 retrieves the initial candidate set and ColPali rescores it, DocRec@5 is preserved at 0.93 but PageRec@5 drops from 0.44 (BGE-M3 + RR) to 0.29. Visual rescoring actively degrades page-level ranking relative to a text-based cross-encoder, likely because the patch embedding similarity used by ColPali is dominated by layout and typography signals rather than the specific numerical content

that distinguishes answer-bearing pages from structurally similar adjacent pages in the same filing.

The two proposed methods close most of the gap to the oracle document bound. The page-then-chunk scorer achieves PageRec@5 of 0.55 and the fine-tuned cross-encoder reranker achieves 0.56, leaving residual gaps of 0.05 and 0.04 points to the oracle document ceiling of 0.60. Both systems achieve BLEU@5 of 0.33, which exceeds the oracle document BLEU@5 (0.25), and the page-then-chunk scorer’s ROUGE-L@5 (0.46) also exceeds the oracle document ROUGE-L@5 (0.42). This super-oracle context quality is not a measurement artefact: the page-then-chunk scorer concentrates chunk retrieval on the 20 highest-scoring pages in the corpus, and when the gold pages fall within that set, the filtered chunk pool is extremely evidence-dense compared with the full within-document pool used by oracle-document retrieval. The fine-tuned reranker, by contrast, achieves its PageRec@5 gain through joint query-passage scoring rather than search-space restriction, and consequently does not exhibit the same ROUGE-L boost because it still draws chunks from the full corpus.

4.2.2. Effect of Retrieval Budget

Figure 4.1 shows how PageRec@ k and DocRec@ k evolve as the retrieval budget k increases from 1 to 20 for selected methods.

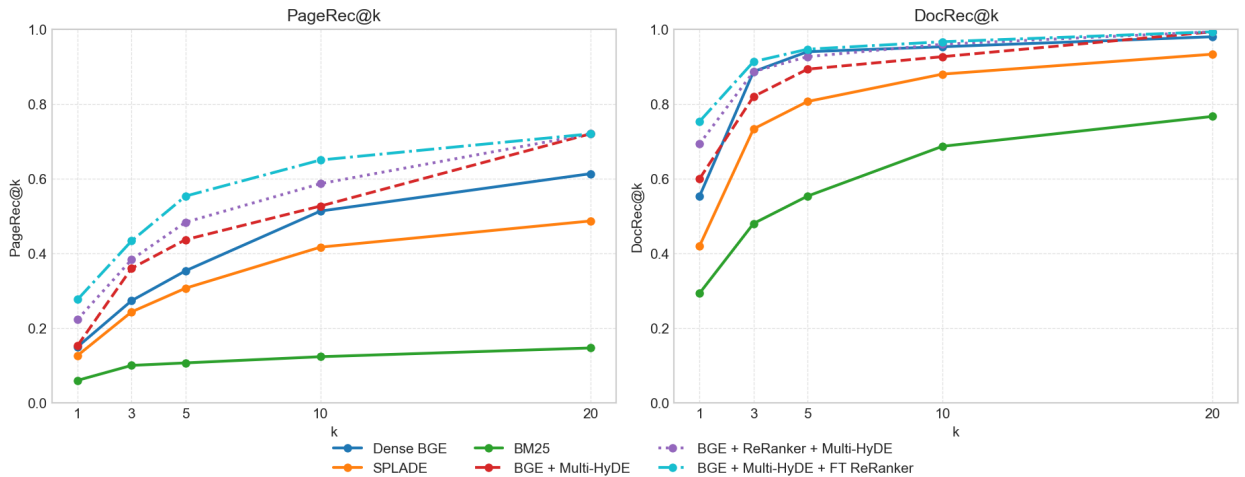


Fig. 4.1. PageRec@ k (left) and DocRec@ k (right) for selected methods at $k \in \{1, 3, 5, 10, 20\}$.

The DocRec@ k curves (right panel) saturate rapidly for dense methods: BGE-M3 reaches 0.95 at $k = 5$ and all dense configurations converge near 0.98–0.99 by $k = 20$. BM25 saturates much later and at a lower ceiling (0.76 at $k = 20$), confirming that the lexical

mismatch is a systemic retrieval failure rather than a budget problem. This fast document-level saturation means that investing additional budget in dense retrieval yields diminishing returns for document discovery, and the diagnostic attention should shift to within-document page selection.

The PageRec@ k curves (left panel) tell a different story. No method saturates within the evaluated budget range, and all curves show sustained growth from $k = 5$ to $k = 20$. For the strongest non-proposed baseline (BGE-M3 + Multi-HyDE + ReRanker), increasing k from 5 to 20 yields a gain of 0.23 points ($0.48 \rightarrow 0.71$). The fine-tuned reranker (BGE-M3 + Multi-HyDE + FT-RR) reaches PageRec@5 of 0.55 at $k = 5$ and 0.72 at $k = 20$, matching the best baseline’s $k = 20$ ceiling at less than a third of the retrieval budget. This efficiency gain has a direct practical consequence: the proposed system delivers high-recall context in the top 5 chunks, passing a smaller, less noisy context window to the generator and reducing the risk of the generator being distracted by off-target content.

The separation between BM25 and all dense methods at $k = 1$ is especially informative. At $k = 1$, BM25 achieves PageRec of only 0.06 while dense BGE-M3 achieves 0.15 and the fine-tuned reranker achieves 0.28, a five-fold improvement. This early-rank quality gap persists at all budgets and is not closed by additional retrieval, confirming that the ranking signal itself, not just the recall ceiling, is fundamentally different across retrieval families.

4.2.3. Performance by Question Type

Table 4.3 decomposes DocRec@5 and PageRec@5 by the three FinanceBench question types for four configurations, together with the oracle document bound. Each question type contains 50 questions. These four configurations were selected to isolate the incremental contribution of each component (reranking, query reformulation, and fine-tuning) relative to the dense baseline.

Table 4.3. DocRec@5 and PageRec@5 at $k = 5$ by question type. MG = metrics-generated; DR = domain-relevant; NG = novel-generated.

Method	MG		DR		NG	
	D@5	P@5	D@5	P@5	D@5	P@5
Dense BGE-M3	0.98	0.49	0.92	0.22	0.92	0.35
BGE-M3 + ReRanker	1.00	0.77	0.86	0.18	0.98	0.38
BGE-M3 + Multi-HyDE + ReRanker	1.00	0.78	0.84	0.29	0.94	0.38
BGE-M3 + Multi-HyDE + FT ReRanker	0.98	0.80	0.90	0.32	0.96	0.54
Oracle document	1.00	0.68	1.00	0.53	1.00	0.58

The three question types exhibit sharply different retrieval profiles, and the relative contribution of each pipeline component varies substantially across them.

On metrics-generated questions, the fine-tuned reranker achieves PageRec@5 of 0.80, exceeding the oracle document bound of 0.68 by a Δ_{page} of 0.12 points. Financial statement pages (income statements, balance sheets, cash flow statements) occupy structurally predictable positions in 10-K filings, and the FinQA training corpus from which the fine-tuned reranker learns is dominated by precisely these page types. The model therefore acquires a strong structural prior toward financial statement pages that the oracle document condition, which applies zero-shot dense chunk retrieval within the gold filing, does not have. This explains the super-oracle result: the fine-tuned reranker’s cross-attention over the full query-page pair captures the semantic alignment between a numerical question and a financial table more precisely than symmetric cosine similarity over chunked passages can. Even the zero-shot cross-encoder reranker (0.77) substantially exceeds oracle document (0.68) on this group, confirming that joint query-page scoring is itself sufficient to capture financial statement structure, independent of fine-tuning. Multi-HyDE adds only 0.01 point on top of the zero-shot reranker ($0.77 \rightarrow 0.78$), suggesting that for structurally predictable evidence, query reformulation provides minimal additional signal once reranking is already present.

On domain-relevant questions, performance drops sharply for all methods. The fine-tuned reranker reaches PageRec@5 of only 0.32, closing 12% of the gap from the best baseline (0.29) to the oracle document bound (0.53). Two factors compound to produce this underperformance. First, domain-relevant questions use generic financial language (e.g. “What is the company’s revenue recognition policy?”) that matches multiple structurally unrelated filing sections simultaneously. Dense retrieval returns pages from multiple sections that happen to share the same keywords, and without knowing which section type the question targets, the reranker cannot reliably surface the right one over the others. Second, and more fundamentally, the FinQA training corpus is almost exclusively composed of financial statement pages and numerical evidence, containing virtually no MD&A narrative, risk factor disclosures, or accounting policy footnotes. The fine-tuned reranker is therefore effectively untrained on the evidence type that domain-relevant questions require, limiting its advantage over the zero-shot baseline to cases where the narrative is concentrated in a single dense-text page. The zero-shot reranker (0.18) actually performs worse than the dense baseline (0.22) and Multi-HyDE (0.29) on this group without query reformulation, because cross-attention scoring over short, generic queries and long narrative pages produces low confidence scores that degrade ranking rather than improving it. Multi-HyDE restores performance by generating a hypothetical passage that approximates the narrative structure of the target page, giving the reranker a richer query representation to score against.

On novel-generated questions, fine-tuning provides the largest absolute gain, PageRec@5 improves from 0.38 (best baseline) to 0.54, closing 80% of the gap to the oracle document bound of 0.58. Novel questions require company-specific reasoning over financial data that is almost always grounded in financial statement pages, even when the question phrasing is indirect or hypothetical. Because the FinQA training distribution covers these page types extensively, the fine-tuned reranker generalises effectively to the implicit evidence structure of novel questions despite the superficial vocabulary mismatch. The zero-shot reranker adds little on this question type ($0.35 \rightarrow 0.38$ over the dense baseline), and Multi-HyDE without fine-tuning yields the same 0.38, suggesting that query reformulation and generic reranking both fail to capture the company-specific numerical grounding that distinguishes the gold page. Fine-tuning is the decisive factor here, and the remaining 20% of the gap to oracle document consists largely of questions requiring multi-passage synthesis or footnote retrieval that the page scorer cannot yet resolve.

4.2.4. Performance by Document Type

Figure 4.2 stratifies PageRec@5 and DocRec@5 by SEC filing type across all evaluated configurations. The benchmark is dominated by 10-K annual reports ($n = 112$), with smaller samples of 10-Q quarterly reports ($n = 15$), 8-K current reports ($n = 9$), and earnings call transcripts ($n = 14$).

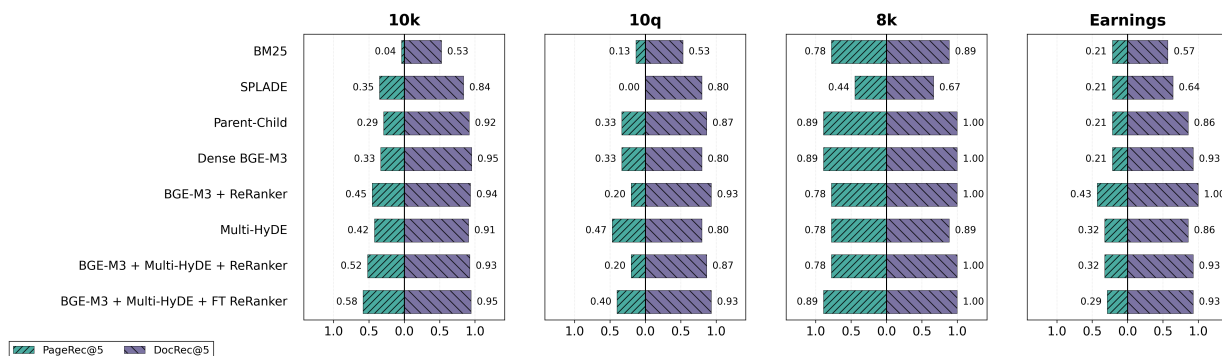


Fig. 4.2. PageRec@5 (left bar, green) and DocRec@5 (right bar, purple) at $k = 5$ by filing type.

On 10-K annual reports, the fine-tuned reranker achieves PageRec@5 of 0.58, exceeding both the best non-proposed baseline (BGE-M3 + Multi-HyDE + ReRanker at 0.52) and the oracle document bound (0.56) for this filing type. As with the metrics-generated question type, the super-oracle result is driven by FinQA’s near-exclusive composition of 10-K evidence pages: the fine-tuned model’s structural prior is calibrated precisely to the page type most prevalent in annual reports, enabling it to rank gold pages above the oracle

document boundary even without document pre-filtering. The dense baseline (0.33) and parent-child (0.29) are substantially weaker, confirming that 10-K filings are long enough that flat dense retrieval is severely penalised by the within-document localisation problem described in Section 4.1.

On 8-K current reports, all methods achieve high PageRec@5, with BM25 reaching 0.78 and the fine-tuned reranker reaching 0.89 (equal to the oracle document bound). This stands in contrast to BM25’s near-zero performance on 10-K filings (0.04). The explanation is structural: 8-K filings are brief current reports, typically containing a single financial exhibit, so the evidence page is easy to locate regardless of retrieval method and lexical matching is sufficient to identify it. SPLADE, however, drops to 0.44 on 8-K despite BM25 reaching 0.78, a counterintuitive reversal. Sparse vocabulary expansion, which helps on longer documents by generating additional matching terms, may over-expand on short 8-K documents, generating spurious matches that push the gold page down in the ranking.

On 10-Q quarterly reports, the fine-tuned reranker (0.40) performs below Multi-HyDE alone (0.47), which is the only filing type where fine-tuning is not strictly beneficial. FinQA is composed exclusively of 10-K annual report filings, so the reranker has no training exposure to quarterly report structure. Unlike earnings calls, which are structurally distinct from all SEC filings, 10-Q reports share the same tabular structure as 10-K reports; the problem is one of temporal content rather than format. Quarterly reports contain interim financial data with different period labels and partial-year figures that the model, trained on annual full-year evidence, scores with lower confidence, leading to degraded ranking even though the page type is structurally familiar. Multi-HyDE alone performs best on this group because generating a hypothetical passage from the query implicitly grounds the retrieval in the question’s period scope without relying on the reranker’s learned annual-report prior.

Finally, we observe out-of-distribution failures on earnings call transcripts. The fine-tuned reranker achieves PageRec@5 of 0.29, which is lower than the zero-shot reranker alone (0.43) and below all methods except BM25 (0.21) and SPLADE (0.21). The fine-tuning actively degrades retrieval relative to the zero-shot baseline by 0.14 points. Earnings call transcripts are conversational, consisting of executive prepared remarks and analyst Q&A exchanges that share no structural features with the tabular regulatory filings that constitute the entirety of the FinQA training corpus. The reranker has learned to assign high scores to pages that contain financial tables, standardised line-item labels, and SEC section headers, but when applied to a conversational transcript, it systematically demotes the correct pages in favour of pages that superficially resemble its learned positive distribution. DocRec@5 is preserved at 0.93 for the fine-tuned reranker on this filing type (equal to the dense baseline), confirming that the document-level retrieval signal is unaffected and the failure is entirely in

within-document page ranking. This is a clear domain-transfer failure where representations learned for standardised regulatory disclosures do not extend to conversational transcripts, and the fine-tuning makes this worse by sharpening the model’s confidence in the wrong page types.

4.2.5. Retrieval Case Analysis

The aggregate and stratified results reveal high-level structural patterns in where and why retrieval fails; the following qualitative analysis provides a closer look at the underlying mechanisms. Three failure modes are distinguished. A total failure occurs when neither the gold document nor any of its gold pages appear in the top- k results. A partial-document failure occurs when the gold document is retrieved but the gold pages are missed. A partial-page failure occurs when the question has multiple gold pages and only a subset are recovered. The subsequent analysis examines each question type through this lens, grounded in concrete examples where they best illustrate the underlying mechanism.

4.2.5.1. Metrics-Generated Questions. Metrics-generated questions require locating specific numerical values in financial tables. The gold evidence almost always resides in a standardised financial statement, making this category the most structurally predictable and the highest-recall category across all methods. Its failure modes are consequently more specific.

The dominant failure mode under dense retrieval alone is temporal confusion, a partial-document failure in which the system retrieves the correct filing structure but from the wrong reporting year. Financial statements are highly standardised across years, the same company’s balance sheets share the same table layout, the same row labels, and the same column structure, differing only in the reported figures and the year header. Dense similarity scores collapse this distinction, treating the two pages as near-duplicates. Consider the following question from FinanceBench:

What is the year-end FY2018 net PP&E for 3M? Answer in USD billions.

Table 4.4 decomposes the retrieval outcome. As can be observed, every structural dimension is correct, the retrieved page is a balance sheet, from the correct company, with the PP&E row and the year-end column. The issue is that the filing year is wrong. The system returns the FY2022 balance sheet instead of FY2018, yielding a generated answer of \$9.17B rather than the correct \$8.70B. The side-by-side pages are reproduced in Appendix C.1.

Component	Table	Document	Row	Column
Ground truth	Correct balance sheet	FY2018	PP&E (net)	Year-end (2018)
Retrieved	Correct balance sheet	FY2022	PP&E (net)	Year-end (2022)

Table 4.4. Error decomposition for a temporal confusion failure. Table type, row, and column are all correct; the only error is the document-level filing year.

Once the cross-encoder reranker is introduced, this failure mode largely disappears. The reranker performs joint scoring over the full query-page pair and attends directly to the year tokens in both the question and the table header, demoting the wrong-year page in favour of the temporally matching filing. This accounts for most of the PageRec@5 gain from 0.49 (dense BGE-M3) to 0.77 (BGE-M3 + ReRanker) on this question type. The residual failures that persist after reranking are lexical gap failures of the same kind described for novel-generated questions in Section 4.2.5.2.

4.2.5.2. Novel-Generated and Domain-Relevant Questions. Novel-generated questions require company-specific reasoning, synthesis across adjacent passages, or computation that cannot be answered by locating a single sentence or table cell. The most consequential failure mode is a lexical gap between the query vocabulary and the vocabulary of the answer-bearing page. The following example illustrates this directly (see Appendix C.2 for the full document pages and additional examples)..

Example 4 | Lexical Gap / Domain Vocabulary Failure

Question. If JPMorgan went bankrupt by the end of Q1 2021 and liquidated all of its assets to pay shareholders, how much could each shareholder receive?

Expected page. p. 5 (reports tangible book value per share (TBVPS) of **\$66.56** as of March 31, 2021).

Retrieved pages. p. 81 (1st), p. 79 (2nd), p. 83 (3rd), p. 77 (4th), p. 85 (5th).

Expected answer. **\$66.56** per share.

Generated answer. *“Equity (assets minus liabilities) equals \$180,714 million. With 4,104,933,895 shares outstanding, each shareholder would receive approximately **\$43.99** per share.”*

The query uses informal language (“liquidate assets per shareholder”) that does not lexically match the financial term *tangible book value per share* reported in the filing. Retrieval

consequently fails to surface the correct capital-metrics page and instead ranks balance sheet pages containing assets and liabilities. While these pages are semantically related, they require additional derivation to produce the answer. The generator performs a from-scratch calculation using retrieved values and produces a numerically plausible but incorrect estimate, rather than reading off the directly reported metric. This failure mode is particularly resistant to reranking because the reranker also lacks a signal connecting “liquidate assets” to “tangible book value”, it would require query expansion or domain glossary alignment to bridge the vocabulary gap at retrieval time.

The fine-tuned reranker improves novel-generated PageRec@5 from 0.38 (best baseline) to 0.54, closing 80% of the gap to the oracle document bound of 0.58. Gains are concentrated in questions whose implicit evidence resides on structurally predictable pages; questions requiring footnote retrieval or multi-passage synthesis remain the hardest subgroup.

4.2.5.3. Annotation Limitation. A recurring source of false retrieval failures is benchmark annotation that designates a single gold page per question even when the text required to answer appears on other pages of the same filing. Because PageRec@5 is computed against the annotated gold set, retrieving an equally informative page that was not annotated counts as a miss, artificially deflating recall while the downstream generator still produces the correct answer. The following example illustrates this directly.

Example 5 | Multi-Location Answer / Annotation Mismatch

Question. What is the quantity of restructuring costs directly outlined in PepsiCo’s income statement for FY2022? If restructuring costs are not explicitly outlined, state 0.

Expected answer. \$411 million.

Generated answer. “*The quantity of restructuring costs directly outlined in PepsiCo’s income statement for FY2022 is **\$411 million.***”

Annotated gold page (Appendix C.5). The restructuring table presents *Total restructuring and impairment charges* of \$411M for FY2022 alongside prior-year comparatives.

Top-1 retrieved page (Appendix C.6), not annotated. The Consolidated Statement of Cash Flows, ranked first by the retrieval system, contains the line *Restructuring and impairment charges* with the identical figure of \$411M as a non-cash adjustment to net income.

Retrieval outcome. p. *CF* (1st, not annotated), p. *GT* (not retrieved within top-5).

The generator produces the correct answer from the retrieved cash-flow page because it carries the identical monetary figure. Yet the evaluation framework records PageRec@5 of zero because the gold annotation points exclusively to the restructuring-table page. This pattern arises whenever a key figure is disclosed in multiple sections of the same filing, which is common for summary metrics such as restructuring charges, depreciation, and tax provisions. It underscores why evaluation must span multiple granularities: page-level recall alone cannot distinguish a genuine retrieval failure from an annotation artefact, and only chunk-level and generative metrics can recover the true system utility in such cases.

4.2.5.4. Failure Mode Summary. Across question types, retrieval failures decompose into four recurring mechanisms. For domain-relevant questions, semantic drift—where generic financial queries match multiple structurally unrelated filing sections—combines with multi-page evidence diffusion and cross-filing boilerplate confounding to leave the majority of the gap to the oracle document bound unresolved; fine-tuning on FinQA provides minimal benefit here because FinQA contains virtually no narrative MD&A or policy disclosure evidence. For metrics-generated questions, temporal confusion between near-duplicate financial statement pages from adjacent filing years is the dominant failure, largely mitigated by the cross-encoder reranker’s attention to year tokens but not entirely eliminated; page-boundary splits and multi-location figure disclosure are secondary contributors. For novel-generated questions, lexical mismatch between informal query phrasing and formal financial terminology causes retrieval to surface semantically related but structurally incorrect pages, and the fine-tuned reranker provides the largest absolute gain precisely because FinQA covers the financial statement evidence types these questions implicitly require. Annotation fragility is a cross-cutting issue that creates systematic underestimates of actual utility detectable only through generative evaluation.

These patterns motivate the generation failure analysis in Section 4.3.5, which examines the residual errors that persist even when retrieval succeeds.

4.3. Generative Evaluation

This section evaluates downstream answer quality produced by the RAG pipeline under three representative retrieval conditions and the two oracle baselines. The primary metrics are NumMatch on the 50 metrics-generated questions, as well as the full 150 question set, and human accuracy over all 150 questions, where human evaluation marks each generated answer as correct or incorrect. ROUGE-L and BERTScore serve as surface-level consistency checks. Section 4.3.1 reports aggregate results and contextualises them against the oracle

bounds established in Section 4.1. Section 4.3.2 examines how answer quality scales with generator size and situates the proposed system within published results on FinanceBench. Sections 4.3.3 and 4.3.4 report breakdowns by question type and document type, incorporating both automatic metrics and human accuracy to give a complete picture of where the system succeeds and where it fails. Section 4.3.5 analyses the residual generation failures that persist even under oracle retrieval conditions.

4.3.1. Overall Answer Quality

Table 4.5 reports downstream answer quality for three retrieval configurations and the two oracle baselines, using Qwen-2.5-7B-Instruct and Llama-3.1-8B-Instruct with the fixed prompt template in Appendix A. The proposed method refers to BGE-M3 dense retrieval combined with Multi-HyDE query reformulation and the domain fine-tuned cross-encoder reranker (Multi-HyDE + FT-RR). NumMatch (m) is evaluated on the 50 metrics-generated questions; NumMatch (all) covers all 150 questions.

LLM	Retrieval	ROUGE-L	BERT-F1	NumMatch (m)	NumMatch (all)
Qwen2.5-7B	Dense BGE-M3	0.13	0.08	0.32	0.33
	Multi-HyDE + RR	0.13	0.06	0.50	0.38
	Multi-HyDE + FT-RR (Ours)	0.14	0.08	0.52	0.42
	Oracle Doc	0.13	0.08	0.44	0.37
	Oracle Page	0.17	0.10	0.70	0.53
Llama3.1-8B	Dense BGE-M3	0.12	0.08	0.36	0.29
	Multi-HyDE + RR	0.13	0.07	0.56	0.33
	Multi-HyDE + FT-RR (Ours)	0.13	0.10	0.52	0.35
	Oracle Doc	0.13	0.08	0.46	0.33
	Oracle Page	0.17	0.13	0.78	0.47

Table 4.5. Generative performance at $k = 5$ for selected baselines, the proposed method (Multi-HyDE + FT-RR), and oracle conditions. NumMatch (m) is evaluated on the 50 metrics-generated questions; NumMatch (all) covers all 150 questions.

The proposed method exceeds the oracle document generation bound on both generators. Multi-HyDE + FT-RR achieves NumMatch (m) of 0.52 with Qwen-2.5-7B, surpassing Oracle Doc (0.44) by 0.08 points, and 0.52 with Llama-3.1-8B against an Oracle Doc of 0.46. This super-oracle result mirrors the retrieval finding in Section 4.2.3: the fine-tuned reranker surfaces the correct evidence pages so reliably that the filtered context passed to the generator is cleaner than what is produced by a brute-force search restricted to the gold document, which still applies zero-shot chunk retrieval and introduces more noise within the filing.

Compared to Dense BGE-M3 (0.32 with Qwen), the proposed method delivers a 0.20-point gain on metrics-generated questions, a 63% relative improvement.

ROUGE-L remains modest across all configurations: even under oracle page conditions it reaches only 0.165, reflecting that the generator frequently paraphrases and reasons over the retrieved evidence rather than extracting text verbatim. ROUGE-L is therefore an insensitive metric for this benchmark, and NumMatch (m) is the primary criterion for metrics-generated questions, with ROUGE-L serving as a surface-form consistency check across question types.

Oracle page NumMatch (m) of 0.70 establishes the generation ceiling: roughly 30% of numeric errors persist even with perfect retrieval, attributable to generation failures rather than retrieval failures. The generation ceiling gap of 0.26 points (0.70 – 0.44) bounds what better retrieval alone can achieve and motivates the failure analysis in Section 4.3.5.

One notable asymmetry between the two generators: Llama-3.1-8B reaches NumMatch (m) of 0.56 with Multi-HyDE + RR, outperforming the Qwen-2.5-7B equivalent (0.50), but the gap reverses under oracle page conditions where Qwen reaches 0.70 against Llama’s 0.78. This suggests that the two models respond differently to improvements in context quality, and that neither dominates the other uniformly across retrieval conditions.

4.3.2. Generator Scaling and Comparison with Prior Work

Table 4.6 examines how answer quality changes as the generator is scaled from 3B to 14B parameters within the Qwen 2.5 model family, with retrieval fixed to the proposed method (Multi-HyDE + FT-RR). Results are shown under two conditions: top- k retrieval, where the proposed pipeline supplies the context, and oracle context, where the gold evidence pages are provided directly. Automatic metrics (NumMatch, ROUGE-L, BERTScore) are reported on the metrics-generated subset; human accuracy is a binary correct/incorrect judgement conducted over all 150 questions under top- k retrieval, providing a broader view of answer quality that covers question types where numeric matching is not applicable.

Table 4.6. Generator performance as a function of model size, with retrieval fixed to Multi-HyDE + FT-RR. NumMatch is on 50 metrics-generated questions; Human Acc. covers all 150.

Metric	Top- k retrieval			FB Global	Oracle context			FB Oracle
	3B	7B	14B	GPT-4	3B	7B	14B	GPT-4
NumMatch	0.40	0.44	0.50	–	0.54	0.66	0.74	–
Human Acc.	55/150 (36.67%)	65/150 (43.33%)	76/150 (50.67%)	29/150 (19.33%)	74/150 (49.33%)	88/150 (58.67%)	109/150 (72.67%)	128/150 (85.33%)

NumMatch scales consistently with model size under both retrieval conditions: from 0.40 to 0.50 under top- k retrieval, and from 0.54 to 0.74 under oracle context. The gains under oracle context are substantially larger, indicating that larger models exploit high-quality evidence more effectively when it is available. This is reflected in the widening gap between oracle-context and top- k NumMatch across model sizes: 0.14 (3B), 0.22 (7B), and 0.24 (14B). Rather than closing the retrieval gap, larger generators amplify it: when the correct evidence is present, larger models use it more accurately; when it is absent or buried among irrelevant passages, the additional capacity provides diminishing return. Retrieval improvements therefore yield super-linear returns as generator capacity increases.

Human accuracy tells the same story but over the full 150 questions. It grows steadily from 36.7% (3B) to 43.3% (7B) to 50.7% (14B) under top- k retrieval, and from 49.3% to 58.7% to 72.7% under oracle context. The oracle-context gains are again larger, confirming that the scaling benefit is not uniform across retrieval conditions. The gap between top- k and oracle human accuracy grows from 12.7 points at 3B to 22.0 points at 14B, mirroring the NumMatch pattern and providing convergent evidence that larger models are more sensitive to retrieval quality across all question types, not just numeric ones.

ROUGE-L and BERTScore remain largely flat across model sizes under top- k retrieval (R-L: 0.138 / 0.139 / 0.135), confirming that scaling improves answer accuracy without substantially changing surface-form generation style.

Table 4.7 contextualises the proposed system within published results on the 150-question FinanceBench open-source subset, using human accuracy for comparability with prior work.

Table 4.7. Comparison with published results on the 150-question FinanceBench subset. Dashes indicate metrics not reported by the original work.

System	Generator	LLM Acc.	Manual Acc.
Islam et al. [2023]	GPT-4-Turbo	–	0.19
Yepes et al. [2024]	GPT-4	0.33	0.37
Setty et al. [2024]	GPT-4o	0.26	–
Wang et al. [2025]	Qwen2.5-72B-Instruct	0.50	0.57
Choe et al. [2025]	GPT-4o	0.50	–
Ours (Multi-HyDE + FT-RR)	Qwen2.5-14B-Instruct	–	0.51

The proposed system with a 14B open-source generator achieves 51% human accuracy, matching FinSage’s LLM-graded score (50%) and narrowing the gap to FinSage’s manual evaluation (57%) to 6 points. FinSage employs a multi-modal preprocessing pipeline, a DPO-trained domain-specialised reranker, and a 72B generator; the proposed system relies on a

fine-tuned bi-encoder trained exclusively on FinQA data and a fixed open-source 14B generator with no proprietary components. The 7B variant (43.3%) already exceeds the closest comparable published system using human evaluation [Yepes et al., 2024] by 6 points, with substantially lower generator capacity. These results confirm that improvements in page-level retrieval account for a large share of the performance gain over the FinanceBench baseline, and that the remaining gap to FinSage is plausibly attributable to their larger generator and DPO-trained reranker, both of which address failure modes identified in Sections 4.2.5.1 and 4.3.5.

4.3.3. Performance by Question Type

Table 4.8 and Figure 4.3 report answer quality by question type for the proposed method (Multi-HyDE + FT-RR) and the dense baseline, using Qwen-2.5-7B-Instruct. Figure 4.3 additionally shows human accuracy for both the 7B and 14B models under top- k retrieval and oracle context, allowing a direct comparison of how model scale and retrieval quality interact across question types.

Retrieval	Model	Q-Type	ROUGE-L	BERT-F1	NumMatch
Dense BGE-M3	Qwen2.5-7B	Metrics	0.01	–	0.32
		Domain	0.17	0.13	–
		Novel	0.21	0.19	–
Multi-HyDE + FT-RR (Ours)	Qwen2.5-7B	Metrics	0.01	–	0.52
		Domain	0.19	0.14	–
		Novel	0.22	0.20	–
Oracle Page	Qwen2.5-7B	Metrics	0.02	–	0.70
		Domain	0.22	0.17	–
		Novel	0.25	0.24	–

Table 4.8. Generative performance by question type (Qwen2.5-7B). NumMatch is only defined for metrics-generated questions.

The NumMatch gains for metrics-generated questions directly track the retrieval gains from Section 4.2.3: the proposed method improves NumMatch (m) from 0.32 to 0.52, a gain of 0.20 points attributable to retrieval improvement, while the oracle page ceiling (0.70) identifies a further 0.18-point headroom that belongs to generation failures analysed in Section 4.3.5. The negative BERT-F1 values for metrics-generated questions across all configurations are expected: short numeric answers have very low lexical overlap with the verbose

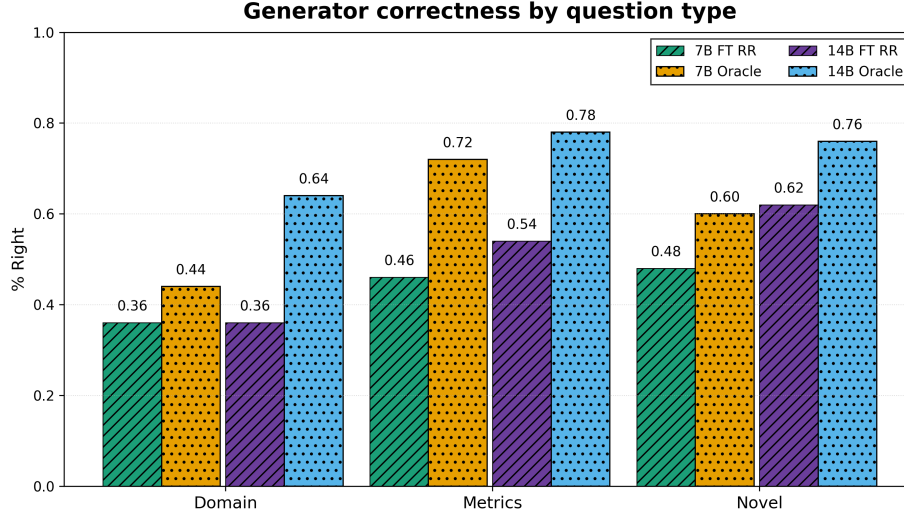


Fig. 4.3. Human accuracy by question type for the 7B and 14B models under top- k retrieval (FT-RR) and oracle context.

gold evidence strings that BERTScore compares against, making it an unreliable metric for this question type and not indicative of a quality failure.

Human accuracy in Figure 4.3 adds important nuance that NumMatch alone does not capture. On metrics-generated questions, the 7B model reaches 46% under top- k retrieval and 72% under oracle context, while the 14B model reaches 54% and 78% respectively. The oracle gain for both models (roughly 26 points) is the largest of any question type, confirming that retrieval quality is the dominant bottleneck here and that both model sizes are capable of correctly answering a substantially larger fraction of these questions given the right context.

Domain-relevant questions show a notably different scaling pattern. The 7B and 14B models achieve identical human accuracy under top- k retrieval (0.36 each), indicating that the retrieval failure is so severe on this question type that model size provides no additional benefit: both models receive essentially the same inadequate context and produce similarly poor answers. Under oracle context, however, the 14B model reaches 0.64 against the 7B model’s 0.44, a 20-point gap that does not appear under top- k retrieval. This reveals that domain-relevant questions are simultaneously a retrieval problem and a generation problem: the 14B model has substantially stronger reasoning over narrative evidence when it is provided correctly, but retrieval failure masks this capacity difference entirely under real operating conditions.

Novel-generated questions show the largest absolute human accuracy gain from model scaling under top- k retrieval: the 14B model (0.62) outperforms the 7B model (0.48) by 14 points, a gap that is larger here than on any other question type. Under oracle context

the gap narrows slightly (0.76 vs. 0.60), suggesting that the 14B model’s advantage on novel questions is partly driven by better reasoning given imperfect context, not only by better use of perfect evidence. The 7B model’s oracle gain for novel questions (0.48 $\rightarrow$ 0.60) is smaller than for metrics questions (0.46 $\rightarrow$ 0.72), consistent with the retrieval analysis showing that novel questions have a smaller remaining gap to the oracle document bound.

For domain-relevant and novel-generated questions, the proposed method also improves ROUGE-L modestly over the dense baseline (domain: 0.17 $\rightarrow$ 0.19; novel: 0.21 $\rightarrow$ 0.22). These small gains are consistent with the retrieval analysis: domain-relevant questions retain a large unresolved retrieval gap, and the generation metrics reflect this difficulty rather than a generator limitation.

4.3.4. Performance by Document Type

Table 4.9 and Figure 4.4 report generative performance by SEC filing type for the proposed method (Multi-HyDE + FT-RR), the dense baseline, and oracle page conditions, using Qwen-2.5-7B-Instruct. Figure 4.4 shows human accuracy for both the 7B and 14B models under top- k retrieval and oracle context, providing a fuller picture of filing-type difficulty than automatic metrics alone.

Retrieval	Doc	ROUGE-L	BERT-F1	NumMatch
Dense BGE-M3	10-K	0.0944	0.0402	0.321
	10-Q	0.1710	0.1309	0.200
	8-K	0.3304	0.3072	0.444
	Earn.	0.2287	0.2462	0.500
Multi-HyDE + FT-RR (Ours)	10-K	0.1095	0.0414	0.429
	10-Q	0.1581	0.1224	0.200
	8-K	0.3347	0.3207	0.444
	Earn.	0.2442	0.2354	0.571
Oracle Page	10-K	0.1322	0.0435	0.545
	10-Q	0.1935	0.1917	0.267
	8-K	0.3979	0.4206	0.444
	Earn.	0.2444	0.2195	0.714

Table 4.9. Generative performance by filing type (Qwen2.5-7B). NumMatch is computed over all questions within each filing-type group; questions with non-numeric reference answers contribute 0 to the average.

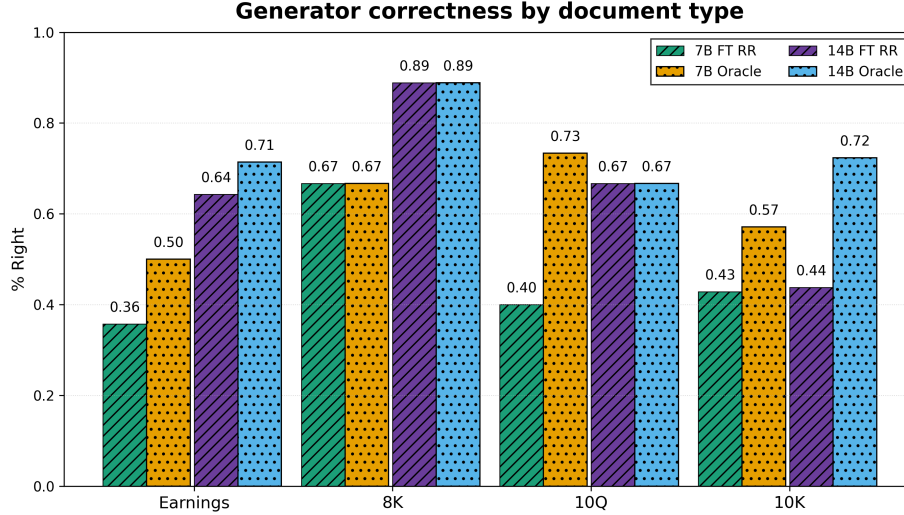


Fig. 4.4. Human accuracy by filing type for the 7B and 14B models under top- k retrieval (FT-RR) and oracle context.

On 10-K annual reports, fine-tuning delivers the largest NumMatch gain across filing types ($0.321 \rightarrow 0.429$), consistent with the retrieval gains in Section 4.2.4. Human accuracy from Figure 4.4 shows 7B at 43% and 14B at 44% under top- k retrieval, nearly identical despite the parameter difference. Under oracle context, however, the 14B model reaches 72% against the 7B’s 57%, a 15-point gap. This pattern is consistent with the question-type finding: under imperfect retrieval both models receive similar noisy context and produce similarly limited answers, but the 14B model exploits clean evidence substantially better when it is available. The oracle page NumMatch ceiling for 10-K (0.545) and the 14B oracle human accuracy (72%) together suggest that further retrieval improvement would yield meaningful gains, particularly for the larger model.

On 8-K current reports, both NumMatch (0.444 across all conditions) and human accuracy tell the same story. The 7B model reaches 67% under both top- k retrieval and oracle context, confirming that evidence localisation is not the binding constraint: the correct page is found regardless of retrieval method, and the remaining errors are generation failures. The 14B model reaches 89% under both conditions, showing that the larger model resolves many of the generation-stage failures that the 7B model cannot, without needing any retrieval improvement to do so.

On 10-Q quarterly reports, human accuracy reveals a striking pattern not visible in NumMatch alone. The 7B model improves substantially from top- k (40%) to oracle (73%), a 33-point gain that is the largest oracle benefit of any filing type for the 7B model, indicating that retrieval is a severe bottleneck for this filing type. The 14B model, by contrast, achieves 67% under both top- k and oracle conditions, showing essentially no benefit from perfect

retrieval. This divergence suggests that the 14B model has a different failure mode on 10-Q filings: it is not bottlenecked by retrieval quality to the same degree as the 7B model, but it also cannot improve beyond 67% even with perfect evidence, pointing to a generation-stage ceiling that model scaling alone does not resolve within the evaluated range.

Earnings call transcripts show the largest model-scaling effect of any filing type under top- k retrieval: the 14B model (64%) outperforms the 7B model (36%) by 28 points, a gap far larger than on any other filing type. This suggests that the 14B model is substantially better at handling the conversational structure of transcripts even when retrieval is imperfect, which the 7B model cannot compensate for. Under oracle context the gap narrows (71% vs. 50%), indicating that some of the 7B model’s difficulty is retrieval-driven, but the 14B model’s advantage persists even with perfect evidence. The NumMatch figures for earnings calls should still be interpreted with caution given the small group size, but the human accuracy figures, which cover all question types in the group, are more reliable and point to a genuine capacity difference on conversational document formats.

4.3.5. Generation Failure Analysis

Under oracle page conditions, NumMatch (m) is 0.70, meaning roughly 30% of numeric errors persist even with perfect retrieval. Similarly, we achieve 72.67% human accuracy in the Oracle framework. These residual failures fall into four patterns.

First, noisy tabular context. When the correct page contains many numbers organised in a dense table, the generator frequently reads off the wrong value even though the right one is present. It may confuse rows or columns that are adjacent in the table, particularly when multiple reporting periods appear side by side. This explains a pattern that appears across several questions: ROUGE-L is reasonable, indicating that the retrieved content is topically correct, but NumMatch is zero because the specific number extracted is wrong.

Multi-step arithmetic failures. Some questions require the generator to combine two or more figures from the retrieved context to compute a final answer, rather than reading a directly stated value. At 7B and 3B scale, errors in this kind of multi-step computation are common even when the correct evidence is provided, explaining the gap between oracle context NumMatch (0.66 for Qwen-2.5-7B) and the 0.70 ceiling. The 14B model narrows this gap substantially (oracle context NumMatch 0.74), confirming that this type of reasoning scales more reliably with model size than surface-form generation does.

Positional bias. When a question requires evidence from two pages simultaneously, both must appear in the retrieved context for the generator to produce the correct answer. When one of the required pages is ranked low and consequently appears at the tail of the context window, the generator tends to overlook it, a behaviour consistent with the well-documented

lost-in-the-middle phenomenon in long-context LLMs [Liu et al., 2024]. Example 6 illustrates this: both required pages are retrieved, but the page containing one of the two required values is ranked last, causing the generator to report that the information is unavailable despite it being present in the context.

Example 6 | Positional Bias — Adobe FY2015 Operating Cash Flow Ratio

Question. What is the FY2015 operating cash flow ratio for Adobe? Operating cash flow ratio is defined as cash from operations divided by total current liabilities. Round to two decimal places.

Expected pages. p. 58 (total current liabilities) and p. 62 (cash from operations).

Retrieved pages. p. **62** (1st), p. 64 (2nd), p. 60 (3rd), p. 71 (4th), p. **58** (5th).

Expected answer. $\$1,469,502 \div \$2,213,556 \approx 0.66$

Generated answer. *“The net cash provided by operating activities for FY2015 is \$1,469,502 thousand. However, the total current liabilities are not directly provided in the given context. We cannot compute the operating cash flow ratio.”*

Both required pages are retrieved, but the page containing total current liabilities (p. 58) is ranked 5th and falls at the tail of the context window. The generator correctly reads the first required value from the top-ranked page but fails to attend to the second value, producing a spurious “not determinable” response despite the evidence being present.

Domain knowledge deficit. Certain questions require the generator to select the correct inputs to a formula from among several plausible candidates that appear on the same page. Without sufficient domain knowledge to distinguish which inputs are appropriate, the generator may apply a formula that is arithmetically valid but conceptually incorrect, producing a wrong answer despite correct retrieval and sound computation. Example 7 illustrates this: retrieval is correct, the arithmetic is correct, but the generator includes an input that the question specification excludes.

Example 7 | Domain Knowledge Deficit — Netflix FY2015 EBITDA Margin

Question. What is the FY2015 unadjusted EBITDA% margin for Netflix? Calculate using unadjusted operating income and D&A from the cash flow statement.

Expected pages. p. 39 (income statement) and p. 41 (cash flow statement).

Retrieved pages. p. 39 (1st), p. 41 (3rd) — retrieval is correct.

Expected answer. 5.4% $(\$62,283 + \$305,826) \div \$6,779,511$

Generated answer. 6.6% $(\$62,283 + \$79,380 + \$305,826) \div \$6,779,511$

The cash flow statement contains two separate line items that could both be interpreted as depreciation and amortisation inputs. The correct answer uses only one of them (\$305,826K); the generator includes both, adding a second component (\$79,380K) that the question specification does not intend. The arithmetic is correct given the generator’s interpretation, but the interpretation itself is wrong. This failure cannot be resolved through better retrieval; it requires the generator to understand the precise definition of the formula being applied.

These four failure patterns collectively explain the 0.30-point gap between oracle page NumMatch (0.70) and perfect generation. Noisy tabular context and domain knowledge deficits are failures of comprehension given correct evidence; positional bias is a failure of attention over the context window; and arithmetic errors are a capacity limitation that diminishes with model scale. All four are generation-stage phenomena that improved retrieval alone cannot resolve.

4.4. Benchmark Extension Results

This section presents initial results on the benchmark extension described in Section 3.6, which combines FinanceBench with a FinQA-derived subset to evaluate retrieval methods in a larger, mixed-document setting. These results should be read as a first evaluation pass rather than a definitive study: several known issues remain and the generation component has not yet been evaluated on the FinQA subset. The primary contribution is to characterise how each retrieval method behaves when the evaluation corpus becomes larger and more diverse, and to identify the directions that require further work.

4.4.1. Extended Corpus Statistics

The benchmark extension constructs a shared global index by combining the original FinanceBench corpus with a PDF-aligned FinQA test subset. The FinQA subset consists of 530 questions drawn from the 1,147-question FinQA test set, filtered to cases where the corresponding PDF filing is available locally. Gold page annotations were derived by fuzzy-matching the FinQA evidence text against extracted PDF page text as described in Section 3.6, with match scores ranging from 0.71 to 1.00 and a mean of 0.88. There is no document overlap between the two components: all retrieval is performed against the global

206-document index, meaning that FinanceBench questions compete against 122 unrelated FinQA filings and vice versa. Table 4.10 summarises the corpus statistics.

Table 4.10. Corpus statistics for the benchmark extension.

Statistic	FinanceBench	FinQA (subset)	Global
Questions	150	530	680
Unique documents	84	122	206
Total pages (est.)	11,948	18,400	30,400
Total chunks	14,808	22,836	37,644
Avg. pages / doc	142	151	147

4.4.2. Retrieval Results

Table 4.11 reports full retrieval metrics for all methods evaluated against the global 206-document index across all 680 questions. These numbers are not directly comparable to Table 4.2, which used the original 84-document index: adding 122 additional distractor documents approximately doubles the candidate pool, and every method scores lower on the FinanceBench subset as a result. The three Hybrid RRF weight variants (25/75, 50/50, 75/25) produced identical results due to a known implementation bug; only the 50/50 result is shown.

Table 4.12 provides a per-corpus breakdown of DocRec@5 and PageRec@5 for a selection of methods that best illustrate the cross-corpus differences. The full per-corpus metrics are reported in Appendix ??.

4.4.3. Cross-Corpus Analysis

Table 4.12 reveals several inversions between the two corpora that are directly informative about what each retrieval method actually learns.

The most striking inversion is BM25, which achieves PageRec@5 of 0.110 on FinanceBench but 0.419 on FinQA, the single best non-oracle result on that corpus. The explanation lies in how the two benchmarks were constructed. FinQA questions were written by annotators working directly from specific tables in filings; the questions reproduce exact company names, fiscal year labels, and table header terms that appear verbatim on the answer page, giving BM25 near-perfect lexical signal. FinanceBench questions are concept-based and rarely share tokens with the answer page, making lexical matching ineffective. Hybrid RRF 50/50 benefits directly from this: incorporating BM25’s strong lexical signal

Table 4.11. Global retrieval performance over all 680 questions in the combined 206-document index at $k = 5$. Bold marks the best non-oracle result per column.

Method	DocRec@5	PageRec@5	BLEU@5	ROUGE-L@5
<i>Sparse</i>				
BM25	0.810	0.351	0.218	0.388
SPLADE	0.793	0.206	0.179	0.316
<i>Dense</i>				
MPNet	0.882	0.252	0.205	0.346
Finance-Investopedia	0.229	0.037	0.044	0.153
Dense BGE-M3	0.931	0.311	0.226	0.370
BGE-M3 + RR	0.932	0.284	0.214	0.351
<i>Hybrid</i>				
RRF 50/50	0.919	0.393	0.269	0.434
<i>Hierarchical and query reformulation</i>				
Parent-Child	0.912	0.325	0.052	0.157
Query Expansion	0.928	0.305	0.225	0.370
BGE-M3 + HyDE	0.844	0.293	0.219	0.361
BGE-M3 + Multi-HyDE	0.854	0.30	0.22	0.362
BGE-M3 + Multi-HyDE + RR	0.894	0.29	0.22	0.356
<i>Proposed</i>				
BGE-M3 + FT-RR	0.94	0.30	0.22	0.36
Multi-HyDE + FT-RR (Ours)	0.92	0.33	0.24	0.38
Oracle document	1.000	0.420	0.279	0.434
Oracle page	0.999	0.999	0.493	0.697

Table 4.12. DocRec@5 and PageRec@5 broken down by corpus subset for selected methods. FB = FinanceBench subset; FQ = FinQA subset; both evaluated within the global index.

Method	FB subset		FinQA subset	
	D@5	P@5	D@5	P@5
BM25	0.533	0.110	0.889	0.419
Dense BGE-M3	0.927	0.343	0.932	0.302
BGE-M3 + RR	0.907	0.427	0.940	0.243
RRF 50/50	0.867	0.283	0.934	0.425
BGE-M3 + Multi-HyDE	0.853	0.447	0.855	0.258
Multi-HyDE + FT-RR (Ours)	0.907	0.462	0.917	0.291
Oracle document	1.000	0.530	1.000	0.389

lifts PageRec@5 to 0.425 on FinQA while its FinanceBench score (0.283) reflects the noise that BM25 introduces on concept-based queries.

The opposite inversion occurs for Multi-HyDE, which is among the strongest methods on FinanceBench (PageRec@5 of 0.447) but falls below the dense baseline on FinQA (0.258). HyDE generates a hypothetical passage that would answer the query. On FinanceBench, this hypothetical resembles the evidence page because questions ask about specific reported values. On FinQA, many questions require numerical computation over a table; the hypothetical answer is a single computed number rather than a passage resembling the source table, so the generated hypothesis actively points the retriever away from the structured evidence.

Dense BGE-M3 shows the smallest performance gap between the two corpora (0.343 vs 0.302), confirming it as the most stable backbone across query styles. The proposed method (Multi-HyDE + FT-RR) is the strongest on the FinanceBench subset (0.462) but does not outperform the plain dense baseline on FinQA (0.291 vs 0.302). The cross-encoder reranker performs even worse on FinQA (BGE-M3 + RR: 0.243), below the dense baseline. Two factors explain this: the cross-encoder was not trained on financial tables and may down-rank table pages whose extracted text does not read like natural language; and the fine-tuned reranker was trained on FinQA evidence formatted as structured text fields from the original JSON annotations, while the evaluation index stores raw PDF-extracted text where tables are linearised differently, creating a format mismatch that limits its benefit on the corpus it was nominally trained on.

Chapter 5

Conclusion

5.1. Summary of Contributions

This thesis investigated retrieval failures in financial question answering over long SEC filings, framing the problem through the lens of within-document retrieval gaps. Four main contributions were made.

The first is an oracle-based retrieval gap analysis framework that decomposes retrieval failures into document discovery errors and within-document localization errors by evaluating retrieval under the restricted candidate sets defined in Equations (3.2.1) and (3.2.2). Applied to FinanceBench, this analysis reveals that standard RAG pipelines achieve reasonable document recall (0.88 to 0.95) but substantially lower page recall (0.07 to 0.46 across baselines), establishing within-document localization as the primary bottleneck.

The second contribution is a comprehensive baseline evaluation reproducing eleven retrieval strategies on a shared multi-document index, providing an apples-to-apples comparison that controls for implementation choices. The evaluation confirms that dense retrieval dominates sparse methods on FinanceBench due to lexical mismatch, and that HyDE and reranking provide complementary improvements that together achieve the strongest non-learned baseline.

The third contribution is the learned page scorer, a domain fine-tuned bi-encoder that explicitly treats pages as an intermediate retrieval level between documents and chunks. Trained via the Multiple Negatives Ranking loss with document-level cross-validation splits, the page scorer achieves 0.55 page recall, a 0.09 improvement over the best baseline, and closes the gap to oracle document performance from 0.14 to 0.05. On metrics-generated questions, the page scorer exceeds the oracle document bound (0.81 versus 0.68), validating that explicit page-level modelling captures structural regularities that direct chunk retrieval misses.

The fourth contribution is the demonstration that retrieval improvements translate directly to generation quality. With Qwen2.5-7B, the strongest fine-tuned retrieval configuration improves metrics-generated numeric match from 0.32 for Dense BGE-M3 to 0.52, and oracle page context raises it further to 0.70. This confirms that better page localization gives the generator more focused numerical context, while also showing that generation errors remain even when retrieval is close to ideal.

5.2. Limitations and Future Work

The evaluation is based on 150 FinanceBench questions dominated by 10-K filings, and the poor performance on earnings call transcripts (0.10 page recall) indicates that the learned page scorer does not generalize across filing types with limited training representation. The page scorer is trained with random in-batch negatives; hard negative mining from pages within the same filing that are topically related but do not contain the answer would provide more informative training signal. Generative results are based on a single language model, and using larger or differently instruction-tuned models may change the relative rankings of retrieval methods. In addition, the generation component was deliberately kept simple: the top- k retrieved chunks are inserted into a fixed in-context prompt, with no generator fine-tuning, no learned context compression, and no task-specific reasoning scaffold. Future work should optimize the generator more directly, for example through instruction tuning on financial QA, stronger context ordering, evidence compression, tool-assisted arithmetic, and larger or more domain-specialized language models.

Promising directions for future work include combining page scoring with query augmentation, reinforcement learning for iterative retrieval that uses generation feedback to refine page selection, incorporating document structure metadata as additional features for the page scorer, extending the oracle analysis and page scorer framework to multi-document questions requiring cross-filing context aggregation, and evaluating hybrid text-visual page scoring that applies ColPali on predominantly tabular pages while using the text-based scorer for narrative sections.

Continuing the benchmark extension is another important direction. In this work, the PDF-grounded extension could only include the FinQA portion of LoFIN/HiREC, because the SEC-QA portion did not provide gold evidence text that could be mapped reliably to PDF pages. Future versions should add richer question-type labels and reasoning tags to the FinQA-aligned data, since many of its questions are metrics-generated or numerical in style. They should also add more diverse question types and document types, including 8-K filings, 10-Q filings, earnings transcripts, proxy statements, and multi-document comparison questions. Synthetic question and data generation could help here, provided that generated

questions are tied to verifiable PDF evidence spans and filtered for answerability. This would make it possible to build a larger PDF-native financial QA benchmark without sacrificing the page-level supervision needed for diagnostic retrieval evaluation.

5.3. Broader Implications

Beyond financial question answering, this work contributes to the broader understanding of retrieval failures in long-document RAG systems. The oracle-based analysis framework is domain-agnostic and can be applied wherever multi-level annotation is available, such as in legal case files, academic papers with section labels, or clinical records with note-type metadata. The consistent finding that within-document localization is the dominant bottleneck challenges the common practice of optimizing retrieval pipelines primarily at the document level and suggests that intermediate retrieval units such as sections and pages deserve more systematic attention. We hope the diagnostic framework and fine-tuned page scorer introduced here provide useful tools for practitioners building reliable RAG systems over long, structured documents.

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Appendix A

Prompt Templates

A.1. Generator Prompt Template

The prompt template used for all generation experiments is reproduced below.

System: You are a financial analyst answering questions based on SEC filings. Use ONLY the provided context. If the context does not contain the answer, say "I cannot determine this from the provided information."

Context: {retrieved_evidence}

Question: {query}

Answer:

A.2. HyDE and Multi-HyDE Prompt Template

System: You are a financial analyst reviewing SEC filings. Given the following question, write a short passage (3-4 sentences) that would directly answer the question, as if excerpted from a company's SEC filing. Include specific financial figures and terminology as they would appear in the filing.

Question: {question}

Passage from annual report:

Appendix B

FinQA Data

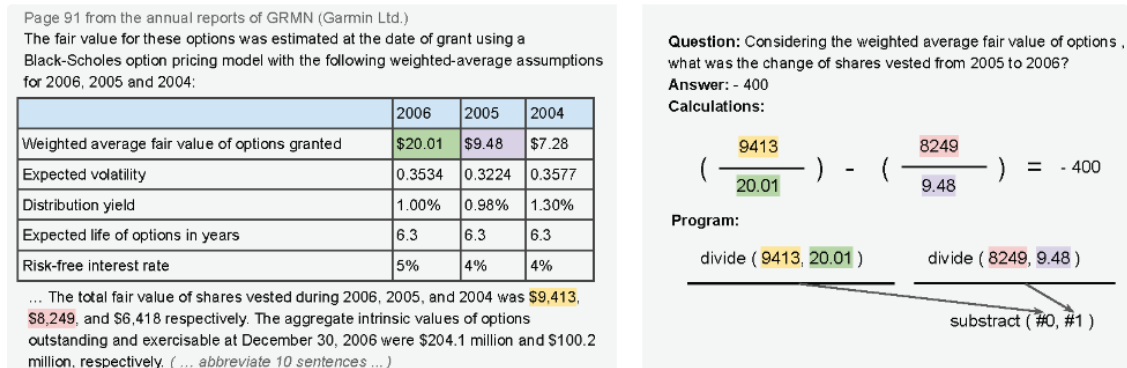


Fig. B.1. FinQA example. From <https://finqasite.github.io/>

Appendix C

Document Page Examples

C.1. Temporal Confusion

3M Company and Subsidiaries
Consolidated Balance Sheet
At December 31

(Dollars in millions, except per share amount)	December 31, 2018	December 31, 2017
Assets		
Current assets		
Cash and cash equivalents	\$ 2,853	\$ 3,053
Marketable securities — current	380	1,076
Accounts receivable — net of allowances of \$95 and \$103	5,020	4,911
Inventories		
Finished goods	2,120	1,915
Work in process	1,292	1,218
Raw materials and supplies	954	901
Total inventories	4,366	4,034
Prepays	741	937
Other current assets	349	266
Total current assets	13,709	14,277
Property, plant and equipment	24,873	24,914
Less: Accumulated depreciation	(16,135)	(16,048)
Property, plant and equipment — net	8,738	8,866
Goodwill	10,051	10,513
Intangible assets — net	2,657	2,936
Other assets	1,345	1,395
Total assets	\$ 36,500	\$ 37,987
Liabilities		
Current liabilities		
Short-term borrowings and current portion of long-term debt	\$ 1,211	\$ 1,853
Accounts payable	2,266	1,945
Accrued payroll	749	870
Accrued income taxes	243	310
Other current liabilities	2,775	2,709
Total current liabilities	7,244	7,687
Long-term debt	13,411	12,096
Pension and postretirement benefits	2,987	3,620
Other liabilities	3,010	2,962
Total liabilities	\$ 26,652	\$ 26,365
Commitments and contingencies (Note 16)		
Equity		
3M Company shareholders' equity:		
Common stock par value, \$.01 par value	\$ 9	\$ 9
Shares outstanding - 2018: 576,575,168		
Shares outstanding - 2017: 594,884,237		
Additional paid-in capital	5,643	5,352
Retained earnings	40,636	39,115
Treasury stock	(29,626)	(25,887)
Accumulated other comprehensive income (loss)	(6,866)	(7,026)
Total 3M Company shareholders' equity	9,796	11,563
Noncontrolling interest	52	59
Total equity	\$ 9,848	\$ 11,622
Total liabilities and equity	\$ 36,500	\$ 37,987

The accompanying Notes to Consolidated Financial Statements are an integral part of this statement.

Fig. C.1. Temporal Confusion: Gold Page

3M Company and Subsidiaries
Consolidated Balance Sheet
At December 31

(Dollars in millions, except per share amount)	2022	2021
Assets		
Current assets		
Cash and cash equivalents	\$ 3,655	\$ 4,564
Marketable securities — current	238	201
Accounts receivable — net of allowances of \$174 and \$189	4,532	4,660
Inventories		
Finished goods	2,497	2,196
Work in process	1,606	1,577
Raw materials and supplies	1,269	1,212
Total inventories	5,372	4,985
Prepays	435	654
Other current assets	456	339
Total current assets	14,688	15,403
Property, plant and equipment	25,998	27,213
Less: Accumulated depreciation	(16,820)	(17,784)
Property, plant and equipment — net	9,178	9,429
Operating lease right of use assets	829	858
Goodwill	12,790	13,486
Intangible assets — net	4,699	5,288
Other assets	4,271	2,608
Total assets	\$ 46,455	\$ 47,072
Liabilities		
Current liabilities		
Short-term borrowings and current portion of long-term debt	\$ 1,938	\$ 1,307
Accounts payable	3,183	2,994
Accrued payroll	692	1,020
Accrued income taxes	259	260
Operating lease liabilities — current	261	263
Other current liabilities	3,190	3,191
Total current liabilities	9,523	9,035
Long-term debt	14,001	16,056
Pension and postretirement benefits	1,966	2,870
Operating lease liabilities	580	591
Other liabilities	5,615	3,403
Total liabilities	31,685	31,955
Commitments and contingencies (Note 16)		
Equity		
3M Company shareholders' equity:		
Common stock par value, \$0.01 par value; 944,033,056 shares issued	9	9
Shares outstanding - December 31, 2022: 549,245,105		
Shares outstanding - December 31, 2021: 571,845,478		
Additional paid-in capital	6,691	6,429
Retained earnings	47,950	45,821
Treasury stock, at cost:	(33,255)	(30,463)
Accumulated other comprehensive income (loss)	(6,673)	(6,750)
Total 3M Company shareholders' equity	14,722	15,046
Noncontrolling interest	48	71
Total equity	14,770	15,117
Total liabilities and equity	\$ 46,455	\$ 47,072

The accompanying Notes to Consolidated Financial Statements are an integral part of this statement.

Fig. C.2. Temporal Confusion: Retrieved Page

C.2. Lexical Gap

Selected capital-related metrics

- The Firm's CET1 capital was \$206 billion, and the Standardized and Advanced CET1 ratios were 13.1% and 13.7%, respectively.
- The Firm's SLR was 6.7%, and without the temporary exclusions of U.S. Treasury securities and deposits at Federal Reserve Banks, 5.5%.
- The Firm grew TBVPS, ending the first quarter of 2021 at \$66.56, up 10% versus the prior year.

Pre-provision profit, ROTCE and TBVPS are non-GAAP financial measures. Refer to Explanation and Reconciliation of the Firm's Use of Non-GAAP Financial Measures on pages 16-17 for a further discussion of each of these measures.

Business segment highlights

Selected business metrics for each of the Firm's four LOBs are presented below for the first quarter of 2021.

CCB ROE 54%	• Average deposits up 32%; client investment assets up 44% • Average loans down 7%; debit and credit card sales volume up 9% • Active mobile customers up 9%
CIB ROE 27%	• Global Investment Banking wallet share of 9.0% in 1Q21 • Total Markets revenue of \$9.1 billion, up 25%, with Fixed Income Markets up 15% and Equity Markets up 47%
CB ROE 19%	• Gross Investment Banking revenue of \$1.1 billion, up 65% • Average loans down 2%; average deposits up 54%
AWM ROE 35%	• Assets under management (AUM) of \$2.8 trillion, up 28% • Average loans up 18%; average deposits up 43%

Refer to the Business Segment Results on pages 18-34 for a detailed discussion of results by business segment.

Credit provided and capital raised

JPMorgan Chase continues to support consumers, businesses and communities around the globe. The Firm provided new and renewed credit and raised capital for wholesale and consumer clients during the first three months of 2021, consisting of:

\$804 billion	Total credit provided and capital raised (including loans and commitments) ^(a)
\$69 billion	Credit for consumers
\$4 billion	Credit for U.S. small businesses
\$300 billion	Credit for corporations
\$417 billion	Capital raised for corporate clients and non-U.S. government entities
\$14 billion	Credit and capital raised for nonprofit and U.S. government entities ^(b)
\$10 billion	Loans under the Small Business Administration's Paycheck Protection Program

(a) Excludes loans under the SBA's PPP.

(b) Includes states, municipalities, hospitals and universities.

Fig. C.3. Lexical Gap: Gold Page

JPMorgan Chase & Co.
Consolidated balance sheets (unaudited)

(in millions, except share data)	March 31, 2021	December 31, 2020
Assets		
Cash and due from banks	\$ 25,397	\$ 24,874
Deposits with banks	685,675	502,735
Federal funds sold and securities purchased under resale agreements (included \$267,613 and \$238,015 at fair value)	272,481	296,284
Securities borrowed (included \$66,277 and \$52,983 at fair value)	179,516	160,635
Trading assets (included assets pledged of \$117,020 and \$130,645)	544,052	503,126
Available-for-sale securities (amortized cost of \$378,756 and \$381,729; included assets pledged of \$64,540 and \$32,227)	379,942	388,178
Held-to-maturity securities (net of allowance for credit losses of \$94 and \$78)	217,452	201,821
Investment securities, net of allowance for credit losses	597,394	589,999
Loans (included \$50,767 and \$44,474 at fair value)	1,011,307	1,012,853
Allowance for loan losses	(23,001)	(28,328)
Loans, net of allowance for loan losses	988,306	984,525
Accrued interest and accounts receivable	114,754	90,503
Premises and equipment	26,926	27,109
Goodwill, MSRs and other intangible assets	54,588	53,428
Other assets ^(a) (included \$50,492 and \$13,827 at fair value and assets pledged of \$37,581 and \$3,739)	200,247	151,539
Total assets^(b)	\$ 3,689,336	\$ 3,384,757
Liabilities		
Deposits (included \$14,107 and \$14,484 at fair value)	\$ 2,278,112	\$ 2,144,257
Federal funds purchased and securities loaned or sold under repurchase agreements (included \$197,834 and \$155,735 at fair value)	304,019	215,209
Short-term borrowings (included \$20,002 and \$16,893 at fair value)	54,978	45,208
Trading liabilities	191,349	170,181
Accounts payable and other liabilities ^(a) (included \$42,824 and \$3,476 at fair value)	285,066	231,285
Beneficial interests issued by consolidated VIEs (included \$193 and \$41 at fair value)	15,671	17,578
Long-term debt (included \$75,693 and \$76,817 at fair value)	279,427	281,685
Total liabilities^(b)	3,408,622	3,105,403
Commitments and contingencies (refer to Notes 22, 23 and 24)		
Stockholders' equity		
Preferred stock (\$1 par value; authorized 200,000,000 shares; issued 3,156,250 and 3,006,250 shares)	31,563	30,063
Common stock (\$1 par value; authorized 9,000,000,000 shares; issued 4,104,933,895 shares)	4,105	4,105
Additional paid-in capital	88,005	88,394
Retained earnings	248,151	236,990
Accumulated other comprehensive income	1,041	7,986
Treasury stock, at cost (1,077,805,783 and 1,055,499,435 shares)	(92,151)	(88,184)
Total stockholders' equity	280,714	279,354
Total liabilities and stockholders' equity	\$ 3,689,336	\$ 3,384,757

(a) Prior-period amounts have been revised to conform with the current presentation. Refer to Note 1 for further information.

(b) The following table presents information on assets and liabilities related to VIEs that are consolidated by the Firm at March 31, 2021, and December 31, 2020. The assets of the consolidated VIEs are used to settle the liabilities of those entities. The holders of the beneficial interests generally do not have recourse to the general credit of JPMorgan Chase. The assets and liabilities in the table below include third-party assets and liabilities of consolidated VIEs and exclude intercompany balances that eliminate in consolidation. Refer to Note 13 for a further discussion.

(in millions)	March 31, 2021	December 31, 2020
Assets		
Trading assets	\$ 1,994	\$ 1,934
Loans	33,509	37,619
All other assets	701	681
Total assets	\$ 36,204	\$ 40,234
Liabilities		
Beneficial interests issued by consolidated VIEs	\$ 15,671	\$ 17,578
All other liabilities	239	233
Total liabilities	\$ 15,910	\$ 17,811

The Notes to Consolidated Financial Statements (unaudited) are an integral part of these statements.

Fig. C.4. Lexical Gap: Retrieved Page

C.3. Annotation Error

A summary of our 2019 Productivity Plan charges is as follows:

	2022	2021	2020
Cost of sales	\$ 33	\$ 29	\$ 30
Selling, general and administrative expenses	347	208	239
Other pension and retiree medical benefits expense	31	10	20
Total restructuring and impairment charges	\$ 411	\$ 247	\$ 289
After-tax amount	\$ 334	\$ 206	\$ 231
Impact on net income attributable to PepsiCo per common share	\$ (0.24)	\$ (0.15)	\$ (0.17)

Fig. C.5. Annotation Limitation: Gold Page

Consolidated Statement of Cash Flows

PepsiCo, Inc. and Subsidiaries

Fiscal years ended December 31, 2022, December 25, 2021 and December 26, 2020

(in millions)

	2022	2021	2020
Operating Activities			
Net income	\$ 8,978	\$ 7,679	\$ 7,175
Depreciation and amortization	2,763	2,710	2,548
Gain associated with the Juice Transaction	(3,321)	—	—
Impairment and other charges	3,618	—	—
Operating lease right-of-use asset amortization	517	505	478
Share-based compensation expense	343	301	264
Restructuring and impairment charges	411	247	289
Cash payments for restructuring charges	(224)	(256)	(255)

Fig. C.6. Annotation Limitation: Retrieved Page

Appendix D

Additional Results

Table D.1 reports retrieval performance of the learned page scorer as a function of the number of pages P selected in Stage 1. Performance plateaus at $P = 20$, which is used as the default throughout this thesis.

Table D.1. Page recall and chunk-level similarity of the learned page scorer as a function of the Stage 1 page budget P .

P	DocRec@5	PageRec@5	BLEU@5	ROUGE-L@5
5	0.92	0.48	0.30	0.44
10	0.95	0.48	0.31	0.44
20	0.95	0.55	0.33	0.46

D.1. Extraction Method Comparison

To select a default PDF extractor for the FinanceBench corpus, we evaluated five tools, PyMuPDF, pdfplumber, PyPDF2, pypdf, and Marker, under identical conditions using the same BGE-M3 dense retrieval setup and reporting PageRec@5 and PageRec@10 across the 150 FinanceBench questions.

Table D.2. Retrieval recall by PDF extraction method on FinanceBench (BGE-M3, no reranking). Higher is better.

Method	Rec@1	Rec@3	Rec@5	Rec@10
pdfplumber	0.147	0.267	0.360	0.500
pypdf	0.153	0.267	0.340	0.487
PyMuPDF	0.147	0.260	0.333	0.493
PyPDF2	0.133	0.267	0.313	0.473
Marker	0.127	0.280	0.327	0.460

Appendix E

Chunking Ablation

To isolate the effect of chunking from other design choices, and to verify the 1024-token configuration adopted from Yepes et al. [2024], we conducted a preliminary ablation holding the retrieval method fixed at BGE-M3 dense retrieval ($k = 5$). We varied chunk size and overlap, and compared them against a page-boundary-aligned variant and a table-aware variant.

Table E.1 reports the available completed results. The table-aware 1024-token configuration successfully retrieves the correct document for most questions (DocRec@5 of 0.913) but yields a much lower page recall at $k = 5$ (0.207) compared to the dense BGE-M3 baseline (0.350) reported in Chapter 4.

This confirms that preserving table-like regions alone is insufficient to solve the page localization bottleneck. While chunking strategy demonstrably impacts text representation, the ablation supports the thesis’s central architectural premise: better page selection via a hierarchical pipeline remains a vastly stronger lever for retrieval performance than hyper-optimizing local chunk boundaries.

Table E.1. Preliminary chunking sweep results. Retrieval metrics are reported at $k = 5$ using BGE-M3.

Strategy	Size	Overlap	DocRec@5	PageRec@5	BLEU@5	ROUGE-L@5
Table-aware	1024	128	0.913	0.207	0.168	0.265
Recursive (Default)	1024	128	0.940	0.350	0.250	0.350